

Phase 0: Mathematical Reboot

A detailed diagnostic and refresher workbook
for the Chebotarev Density Theorem study course

Proof language, algebra, Galois theory, real analysis, and complex analysis

Design principle. This edition is built to help you *recover definitions and the intuition behind them*, not merely to test whether you remember how to solve exercises. Every central definition is paired with: (i) why mathematicians use it, (ii) a mental model, (iii) examples and nonexamples, and (iv) a derivation or proof showing how it enters later mathematics.

How to use this workbook

Phase 0 has one job: restore enough mathematical fluency that later material can be learned for its *mathematical content* rather than being blocked by forgotten language, notation, or standard arguments. It is deliberately selective. You are not trying to redo an MSc.

This revised edition uses a three-pass method.

1. **Recall before recognition.** Before reading a definition, cover the page and try to state it. Recognition (“yes, that looks familiar”) is much weaker than recall.
2. **Reconnect definition to purpose.** After reading a definition, answer: *what problem does this definition solve?* and *what would go wrong with a weaker definition?*
3. **Reconstruct.** Close the book and reproduce one proof or worked example. The goal is not photographic memory; it is to recover the chain of reasons.

Memory hook: A useful rule

If you can state a definition but cannot produce an example and a nonexample, you probably do not yet own the definition. If you can follow a proof but cannot say what the decisive idea is, you probably do not yet own the proof.

Recommended order

1. Take the baseline diagnostic closed-book. Mark each problem C (confident), U (uncertain), or N (no workable approach).
2. Read Chapters 2–6, spending more time on the topics your diagnostic exposed.
3. At the end of each section, do the short *definition recall drill* before exercises.
4. Attempt exercises before consulting Part II, which contains complete solutions.
5. Take the final readiness test closed-book.
6. Use the rubric to decide whether to move to Phase 1 or repair one or two specific topics.

What “ready for Phase 1” means

You are ready when you can usually do four things: state the central definitions accurately; explain why they are natural; reconstruct representative proofs; and calculate standard examples without constant reference to notes. You do *not* need to remember every theorem name or technical trick.

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Chapter 1

Baseline diagnostic: closed book

Do this chapter before reading the refreshers. The purpose is to distinguish *rust* from a genuine gap. Rust often feels like “I know what should happen, but I cannot quite execute.” A genuine gap feels like “I do not know what object or theorem is relevant.” Both are normal after time away from mathematics, but they call for different kinds of review.

For each problem, first write C (confident), U (uncertain), or N (no workable approach). Then attempt it without notes. Do not spend more than about 15 minutes on any one problem.

1.1 Algebra and Galois theory

Exercise: D1: quotient field criterion

Prove that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

Exercise: D2: Chinese remainder theorem in action

Find the unique residue class $x \pmod{105}$ satisfying

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 4 \pmod{7}.$$

Then state the ring-theoretic Chinese remainder theorem for pairwise comaximal ideals.

Exercise: D3: homomorphism and quotient

Let $\varphi : G \rightarrow H$ be a group homomorphism. Prove that $\ker \varphi \triangleleft G$ and construct explicitly an isomorphism

$$G / \ker \varphi \cong \operatorname{im} \varphi.$$

Exercise: D4: group action

Let S_3 act on $\{1, 2, 3\}$ naturally. Determine the orbit and stabilizer of 1 and verify orbit–stabilizer. Then describe the conjugacy classes of S_3 .

Exercise: D5: Sylow reasoning

Let G be a group of order 21. Show that the Sylow 7-subgroup is normal. What can you say about the number of Sylow 3-subgroups?

Exercise: D6: prime and maximal ideals

Let R be a commutative ring with 1 and $I \subsetneq R$ an ideal. Prove:

- (a) I is prime if and only if R/I is an integral domain;
- (b) I is maximal if and only if R/I is a field.

Exercise: D7: minimal polynomial

Let $\alpha = \sqrt{2} + \sqrt{3}$. Find the minimal polynomial of α over $\mathbb{Q}$ and determine $[\mathbb{Q}(\alpha) : \mathbb{Q}]$.

Exercise: D8: finite fields and Frobenius

Let $q = p^r$. Prove that

$$F : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_{q^n}, \quad F(x) = x^q$$

is an automorphism fixing $\mathbb{F}_q$. Show that $F^n = \text{id}$.

Exercise: D9: Galois correspondence

Let $L/\mathbb{Q}$ be the splitting field of $x^3 - 2$. Determine $[L : \mathbb{Q}]$ and $\text{Gal}(L/\mathbb{Q})$ up to isomorphism. Identify the fixed field of the subgroup generated by complex conjugation.

1.2 Analysis

Exercise: D10: series and comparison

Classify each series as absolutely convergent, conditionally convergent, or divergent:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}, \quad \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}, \quad \sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

Justify each answer.

Exercise: D11: pointwise versus uniform convergence

On $[0, 1]$, let $f_n(x) = x^n$. Find the pointwise limit. Does f_n converge uniformly? Prove your answer from the definition or an equivalent criterion.

Exercise: D12: uniform convergence and integration

Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are continuous and $f_n \rightarrow f$ uniformly. Prove

$$\int_a^b f_n(x) dx \rightarrow \int_a^b f(x) dx.$$

Exercise: D13: radius of convergence

Find the radius of convergence of

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n.$$

What happens on the boundary $|z| = R$?

Exercise: D14: Cauchy integral formula

Evaluate, for positively oriented $|z| = 2$,

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz.$$

State the theorem you use and its hypotheses.

Exercise: D15: residues

Compute

$$\int_{|z|=3} \frac{z^2 + 1}{z(z-1)(z-2)} dz$$

with positive orientation.

1.3 Proof fluency

Exercise: D16: infinitely many primes

Prove that there are infinitely many prime numbers. Then identify precisely where contradiction enters the proof.

Exercise: D17: root bound for polynomials

Let K be a field and $0 \neq f \in K[x]$ have degree d . Prove that f has at most d roots in K .

Exercise: D18: compactness-style proof

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and suppose $f(x) > 0$ for every $x \in [0, 1]$. Prove that there exists $m > 0$ such that $f(x) \geq m$ for every $x \in [0, 1]$. State exactly where compactness is used.

1.4 Diagnostic interpretation

Do not score by raw percentage alone. Use the following categories.

Pattern	Interpretation
Many C answers, execution errors	Mostly rust. Read refreshers quickly and emphasize reconstruction.
Many U answers but good partial starts	Definitions are retrievable but not fluent. Spend time on memory hooks and examples.
Many N answers in one subject only	Localized gap. Repair that subject; do not restart all mathematics.
Many N answers across subjects	Use this workbook slowly. Phase 0 may take several weeks rather than ten days.
Strong calculations, weak proof explanations	Rebuild proof language deliberately; later algebraic number theory will demand it.
Strong algebra, weak analysis	Do not panic: Phase 1 is algebraic. Still repair uniform convergence and complex analysis before later analytic phases.

Chapter 2

Proof language and proof reconstruction

The fastest way to feel rusty in mathematics is to remember the “big ideas” while forgetting the logical grammar in which those ideas are expressed. This chapter reactivates that grammar. The aim is not elementary logic for its own sake; it is to make later definitions and proofs feel readable again.

2.1 Statements, predicates, and quantifiers

Definition: statement and predicate

A *statement* is a sentence that is either true or false. A *predicate* $P(x)$ is a statement containing a variable; it becomes a definite statement after x is specified or quantified.

Examples: “7 is prime” is a statement. “ x is prime” is a predicate on integers.

Intuition

A predicate is a mathematical property waiting for an input. A quantifier tells you how broadly that property is being asserted.

The universal and existential quantifiers are

$$\forall x \in X P(x), \quad \exists x \in X P(x).$$

The first says every permitted input passes the test; the second says at least one input does.

Negation changes both the quantifier and the predicate:

$$\neg(\forall x P(x)) \iff \exists x \neg P(x), \quad \neg(\exists x P(x)) \iff \forall x \neg P(x).$$

This is not merely formal symbolism. It is exactly why a single counterexample disproves a universal statement.

Worked example: Negating a convergence statement

The statement $a_n \rightarrow L$ means

$$\forall \varepsilon > 0 \exists N \forall n \geq N : |a_n - L| < \varepsilon.$$

Negating it carefully gives

$$\exists \varepsilon_0 > 0 \forall N \exists n \geq N : |a_n - L| \geq \varepsilon_0.$$

Notice how every quantifier flips. Intuitively, nonconvergence means there is some fixed tolerance ε_0 that the sequence violates arbitrarily far out.

2.2 Implication, converse, contrapositive

Definition: implication

The implication $P \Rightarrow Q$ asserts that whenever P is true, Q must be true. P is a *sufficient* condition for Q , while Q is a *necessary* condition for P .

The converse is $Q \Rightarrow P$ and is logically independent. The contrapositive is

$$\neg Q \Rightarrow \neg P,$$

and is logically equivalent to the original implication.

Memory hook

“Sufficient” points forward: P is enough to force Q . “Necessary” points backward: if P happened, Q had to be present.

A biconditional $P \Leftrightarrow Q$ means both directions. In proofs, separate them explicitly unless one direction is immediate.

2.3 Four proof patterns to reactivate

2.3.1 Direct proof

To prove $P \Rightarrow Q$, assume P and transform the information until Q follows.

Worked example: Even squares

Suppose n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$. Hence

$$n^2 = 4k^2 = 2(2k^2),$$

so n^2 is even. The decisive move is not the algebra; it is unpacking the definition of “even.”

2.3.2 Contrapositive

Sometimes $\neg Q$ has a more usable algebraic form than P .

For instance, to prove “if n^2 is even, then n is even,” it is convenient to prove: if n is odd, then n^2 is odd.

2.3.3 Contradiction

To prove P , assume $\neg P$ and derive an impossibility. Contradiction is especially natural for statements of impossibility or infinitude.

Intuition

Contradiction temporarily enters the mathematical world in which your desired theorem is false. You then show that world cannot be internally consistent with already accepted facts.

2.3.4 Induction

To prove $P(n)$ for all integers $n \geq n_0$:

1. establish $P(n_0)$;
2. assume $P(k)$ for an arbitrary $k \geq n_0$;
3. derive $P(k + 1)$.

The induction hypothesis is not “assuming what you want to prove.” It assumes one instance in order to establish the next.

2.4 Proving equality

There are several common forms of equality and each has a standard proof strategy.

2.4.1 Set equality

To show $A = B$, prove both inclusions:

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

The reason is definitional: two sets are equal precisely when they have the same elements.

2.4.2 Equality of functions

To show $f = g$ as maps $X \rightarrow Y$, prove $f(x) = g(x)$ for every $x \in X$.

2.4.3 Equality of algebraic objects

Often use a universal property or two containments. For ideals, for example, $I = J$ is usually proved by $I \subseteq J$ and $J \subseteq I$.

2.5 Existence and uniqueness

A statement “there exists a unique x such that $P(x)$ ” contains two claims:

1. **existence:** construct or prove at least one x satisfying P ;
2. **uniqueness:** show any two x, y satisfying P must be equal.

These steps often use completely different ideas.

Worked example: Uniqueness of an identity element

Suppose e and e' are both identity elements in a group. Since e is an identity, $ee' = e'$. Since e' is an identity, $ee' = e$. Hence $e = e'$. The proof works by making the two candidates interact in one expression.

2.6 Definitions are compressed theorems

A graduate-level definition is rarely arbitrary. Usually each clause prevents a specific pathology.

For example, a subgroup $H \leq G$ must be closed under multiplication and inverse and contain the identity. But these conditions are related: the one-step subgroup test says a nonempty subset H is a subgroup if

$$a, b \in H \implies ab^{-1} \in H.$$

Why? Setting $a = b$ gives $e = aa^{-1} \in H$; setting $a = e$ gives $b^{-1} \in H$; then $ab = a(b^{-1})^{-1} \in H$.

This kind of derivation is worth practicing because it recovers the internal logic of a definition rather than treating it as a list to memorize.

2.7 Proof reconstruction rather than rereading

When refreshing a theorem, use the following four-step routine:

1. State the theorem from memory, including hypotheses.
2. Write one sentence naming the decisive idea.
3. Reconstruct the proof without looking.
4. Compare with the reference and annotate only the missing transitions.

Memory hook: Proof skeleton

For every important proof, try to retain three things: *input*, *key move*, *output*. Technical details can often be regenerated if those three are stable.

2.8 Definition recall drill

Without looking back, write down:

1. the formal negation of $\forall x \exists y P(x, y)$;
2. the difference between converse and contrapositive;
3. what must be proved for “there exists a unique x ”;
4. how to prove equality of sets;
5. the formal definition of sequence convergence.

Then check your answers against the preceding sections.

2.9 Exercises

Exercise: P1: negating uniform convergence

Let $f_n, f : X \rightarrow \mathbb{R}$. Write the definition of uniform convergence $f_n \rightarrow f$, then negate it completely and correctly.

Exercise: P2: equality by two inclusions

For sets A, B, C , prove

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

by two inclusions.

Exercise: P3: necessary versus sufficient

For integers n , consider the statements P : “ $12 \mid n$ ” and Q : “ $6 \mid n$.” Which is necessary and which is sufficient for the other? Is the converse true? Give a counterexample where appropriate.

Exercise: P4: uniqueness pattern

Prove that inverses in a group are unique.

Exercise: P5: contrapositive

Prove: if n^2 is odd, then n is odd. Write both the direct statement and its contrapositive before choosing a proof.

Exercise: P6: induction

Prove for $n \geq 1$ that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

After the proof, identify exactly where the induction hypothesis is used.

Exercise: P7: hypothesis audit

The statement “every continuous function on an interval attains a maximum” is false as written. Give a counterexample. Then repair the statement by adding a standard hypothesis and state the resulting theorem precisely.

Exercise: P8: proof skeleton

Write a three-line proof skeleton (input, key move, output) for Euclid’s theorem that there are infinitely many primes. Do not write the full proof yet.

Mastery gate: Proof-language gate

Move on when you can translate a definition with nested quantifiers into plain language and back; negate it reliably; distinguish converse from contrapositive; and reconstruct a short proof from its key idea rather than from memorized wording.

Chapter 3

Algebra reboot I: groups, actions, rings, and quotients

This chapter is designed around a single principle: abstract algebra becomes memorable when each definition is tied to the structural problem it solves. Groups formalize reversible symmetry; quotient constructions identify objects that differ only by something we choose to regard as negligible; ideals are exactly the subsets that make ring quotients possible.

3.1 Groups: reversible structure

Definition: group

A *group* is a set G with a binary operation $(g, h) \mapsto gh$ such that:

1. $(gh)k = g(hk)$ for all $g, h, k \in G$;
2. there exists $e \in G$ with $eg = ge = g$ for every g ;
3. for every $g \in G$ there exists $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

If $gh = hg$ for all g, h , the group is *abelian*.

Intuition: Why these axioms?

A group models operations that can be composed and undone. Associativity says that a sequence of operations has an unambiguous result; the identity is “do nothing”; inverses allow every operation to be reversed. Commutativity is *not* required because many symmetries depend on order.

Examples include $(\mathbb{Z}, +)$, $(\mathbb{Q}^\times, \cdot)$, permutation groups S_n , and matrix groups such as $\mathrm{GL}_n(\mathbb{R})$. Nonexamples help: $(\mathbb{Z}, \cdot)$ is not a group because most integers have no multiplicative inverse in $\mathbb{Z}$.

3.1.1 Subgroups

Definition: subgroup

A subset $H \subseteq G$ is a *subgroup*, written $H \leq G$, if it is itself a group under the operation inherited from G .

The most efficient test is the following.

Proposition 3.1 (one-step subgroup test). *A nonempty subset $H \subseteq G$ is a subgroup if and only if*

$$a, b \in H \implies ab^{-1} \in H.$$

Proof. If H is a subgroup, it is closed under inverses and products, so the condition is immediate. Conversely suppose $H \neq \emptyset$ and the condition holds. Choose $a \in H$. Then $aa^{-1} = e \in H$. With $e, b \in H$, the condition gives $eb^{-1} = b^{-1} \in H$. Finally for $a, b \in H$, we now know $b^{-1} \in H$, so applying the condition to a and b^{-1} gives $a(b^{-1})^{-1} = ab \in H$. Thus H contains the identity, inverses, and products. Associativity is inherited from G . $\square$

Memory hook

A subgroup is a collection of symmetries that is closed under “do one, then undo another.” The one-step test packages multiplication and inversion into one expression.

3.2 Homomorphisms: structure-preserving maps

Definition: group homomorphism

A map $\varphi : G \rightarrow H$ is a group homomorphism if

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$$

for all $g_1, g_2 \in G$.

Intuition

A homomorphism may forget information, but it must respect the operation. It is the right notion of “same algebraic behavior, possibly at lower resolution.”

Two subsets summarize what is lost and what survives:

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\}, \quad \text{im } \varphi = \{\varphi(g) : g \in G\}.$$

The kernel consists exactly of elements made invisible by the map.

Proposition 3.2. *For a group homomorphism $\varphi : G \rightarrow H$:*

1. $\varphi(e_G) = e_H$;
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$;
3. $\ker \varphi \triangleleft G$;
4. $\text{im } \varphi \leq H$.

Proof. Since $\varphi(e) = \varphi(e^2) = \varphi(e)^2$, multiply by $\varphi(e)^{-1}$ to get $\varphi(e) = e$. Then

$$e = \varphi(e) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}),$$

so $\varphi(g^{-1}) = \varphi(g)^{-1}$. The subgroup property for image and kernel follows from the subgroup test. For normality, if $k \in \ker \varphi$ and $g \in G$, then

$$\varphi(gkg^{-1}) = \varphi(g)e\varphi(g)^{-1} = e,$$

so $gkg^{-1} \in \ker \varphi$. $\square$

3.3 Normal subgroups and quotient groups

Definition: normal subgroup

A subgroup $N \leq G$ is *normal*, written $N \triangleleft G$, if

$$gNg^{-1} = N \quad \text{for every } g \in G.$$

Equivalently, $gN = Ng$ for every g .

Why normality? We would like to multiply cosets by

$$(gN)(hN) := (gh)N.$$

This operation is meaningful only if it does not depend on the chosen representatives. Suppose $g' = gn_1$ and $h' = hn_2$. Then

$$g'h' = gn_1hn_2 = gh(h^{-1}n_1h)n_2.$$

For this to lie in the same coset ghN , we need $h^{-1}n_1h \in N$. That is exactly normality.

Intuition: What a quotient does

Passing from G to G/N declares two elements equivalent whenever they differ by an element of N . Thus N becomes invisible. A quotient is not merely “dividing a group by a subgroup”; it is systematically forgetting a chosen normal part of the structure.

3.3.1 First isomorphism theorem

Theorem 3.3 (First isomorphism theorem for groups). *If $\varphi : G \rightarrow H$ is a homomorphism, then*

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

Proof. Define

$$\bar{\varphi} : G/\ker \varphi \rightarrow \operatorname{im} \varphi, \quad \bar{\varphi}(g\ker \varphi) = \varphi(g).$$

The key point is well-definedness. If $g\ker \varphi = h\ker \varphi$, then $h^{-1}g \in \ker \varphi$, so

$$e = \varphi(h^{-1}g) = \varphi(h)^{-1}\varphi(g),$$

hence $\varphi(g) = \varphi(h)$. Thus the value depends only on the coset. The map is a homomorphism and is surjective onto $\operatorname{im} \varphi$. If $\bar{\varphi}(g\ker \varphi) = e$, then $g \in \ker \varphi$, so the coset is the identity coset. Hence $\bar{\varphi}$ is injective. $\square$

Memory hook

A homomorphism forgets exactly its kernel. After quotienting out that forgotten information, what remains is precisely the image.

3.4 Group actions: groups acting as symmetries

Definition: group action

A left action of G on a set X is a map $G \times X \rightarrow X$, written $(g, x) \mapsto g \cdot x$, satisfying

$$e \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x).$$

An action is equivalent to a homomorphism

$$G \rightarrow \text{Sym}(X),$$

where $\text{Sym}(X)$ is the group of permutations of X . This is why actions are the right language for symmetry: every group element literally becomes a permutation of the objects being studied.

Definition: orbit and stabilizer

For $x \in X$,

$$\text{Orb}(x) = \{g \cdot x : g \in G\}, \quad \text{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

The orbit records where x can move; the stabilizer records which symmetries leave x fixed.

Theorem 3.4 (Orbit–stabilizer). *If G is finite, then*

$$|G| = |\text{Orb}(x)| |\text{Stab}(x)|.$$

More generally, $|\text{Orb}(x)| = [G : \text{Stab}(x)]$.

Proof. Define a map from left cosets $G/\text{Stab}(x)$ to the orbit by

$$g \text{Stab}(x) \mapsto g \cdot x.$$

It is well-defined because $g \text{Stab}(x) = h \text{Stab}(x)$ implies $h^{-1}g \in \text{Stab}(x)$, so $(h^{-1}g) \cdot x = x$ and therefore $g \cdot x = h \cdot x$. It is clearly surjective. If $g \cdot x = h \cdot x$, then $h^{-1}g \cdot x = x$, so $h^{-1}g \in \text{Stab}(x)$ and the cosets agree. Hence the map is a bijection. $\square$

3.4.1 Conjugation

Every group acts on itself by conjugation:

$$g \cdot x = gxg^{-1}.$$

The resulting orbits are the *conjugacy classes*.

Intuition: Why conjugacy matters later

Conjugate elements describe the same group-theoretic behavior viewed from different coordinates. In Galois theory, changing a prime above a fixed prime changes a Frobenius element by conjugation. This is why Chebotarev is naturally stated for conjugacy classes rather than single elements.

3.5 Sylow theorems: what to retain

If $|G| = p^n m$ with $p \nmid m$, a Sylow p -subgroup has order p^n . The Sylow theorems say:

1. such subgroups exist;

2. all Sylow p -subgroups are conjugate;
3. the number n_p of Sylow p -subgroups satisfies

$$n_p \equiv 1 \pmod{p}, \quad n_p \mid m.$$

The practical memory hook is: *divides the complementary factor and is 1 modulo p* . If these constraints force $n_p = 1$, that Sylow subgroup is unique and therefore normal.

3.6 Rings and ideals

Definition: commutative ring with identity

A commutative ring R is an abelian group under addition, equipped with an associative, commutative multiplication distributing over addition, with multiplicative identity 1.

A ring allows addition, subtraction, and multiplication; division may fail. Much of commutative algebra studies how badly division and factorization fail and what structural substitutes remain.

Definition: ideal

An additive subgroup $I \subseteq R$ is an *ideal* if

$$r \in R, a \in I \implies ra \in I.$$

Intuition: Why ideals, not merely subrings?

An ideal is a subset stable under multiplication by arbitrary ambient elements. That extra absorption property is exactly what is needed for multiplication of residue classes to be well-defined in R/I .

Indeed, if $a \equiv a' \pmod{I}$ and $b \equiv b' \pmod{I}$, then $a - a', b - b' \in I$. We need

$$ab - a'b' = a(b - b') + b'(a - a') \in I,$$

and this uses absorption by arbitrary ring elements.

3.7 Quotient rings

For an ideal $I \triangleleft R$, define

$$R/I = \{a + I : a \in R\}$$

with $(a + I) + (b + I) = (a + b) + I$ and $(a + I)(b + I) = ab + I$.

The quotient map

$$\pi : R \rightarrow R/I, \quad a \mapsto a + I$$

has kernel I . Thus ideals are precisely the kernels of ring homomorphisms (for commutative rings with 1, with the usual conventions about unital maps).

3.8 Prime and maximal ideals

Definition: prime ideal

A proper ideal $\mathfrak{p} \subsetneq R$ is *prime* if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

Intuition

Prime ideals are ideals for which “zero modulo $\mathfrak{p}$ ” behaves like zero in an integral domain: a product can vanish only if one factor vanishes.

Proposition 3.5. *A proper ideal $\mathfrak{p}$ is prime if and only if $R/\mathfrak{p}$ is an integral domain.*

Proof. In $R/\mathfrak{p}$,

$$(a + \mathfrak{p})(b + \mathfrak{p}) = 0 + \mathfrak{p}$$

if and only if $ab \in \mathfrak{p}$. The quotient has no zero divisors exactly when this implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, equivalently one residue class is zero. $\square$

Definition: maximal ideal

A proper ideal $\mathfrak{m} \subsetneq R$ is *maximal* if there is no ideal I with

$$\mathfrak{m} \subsetneq I \subsetneq R.$$

Proposition 3.6. *A proper ideal $\mathfrak{m}$ is maximal if and only if $R/\mathfrak{m}$ is a field.*

Proof. Suppose $\mathfrak{m}$ is maximal. Let $a + \mathfrak{m} \neq 0$, so $a \notin \mathfrak{m}$. The ideal $\mathfrak{m} + (a)$ strictly contains $\mathfrak{m}$, hence by maximality equals R . Thus

$$1 = m + ra$$

for some $m \in \mathfrak{m}$, $r \in R$. Modulo $\mathfrak{m}$, this says $(r + \mathfrak{m})(a + \mathfrak{m}) = 1 + \mathfrak{m}$, so every nonzero class is invertible.

Conversely, suppose $R/\mathfrak{m}$ is a field and $\mathfrak{m} \subsetneq I \subsetneq R$. Then $I/\mathfrak{m}$ is an ideal in the field $R/\mathfrak{m}$, so it is either 0 or all of $R/\mathfrak{m}$. Thus $I = \mathfrak{m}$ or $I = R$. $\square$

Every maximal ideal is prime because every field is an integral domain. The converse need not hold; for example (0) is prime but not maximal in $\mathbb{Z}$.

3.9 The Chinese remainder theorem

Definition: comaximal ideals

Ideals $I, J \subseteq R$ are *comaximal* if

$$I + J = R.$$

Equivalently, there exist $a \in I$ and $b \in J$ with $a + b = 1$.

Theorem 3.7 (Chinese remainder theorem, two ideals). *If $I + J = R$, then*

$$R/(I \cap J) \cong R/I \times R/J.$$

Moreover $I \cap J = IJ$, so

$$R/IJ \cong R/I \times R/J.$$

Proof. Consider

$$\phi : R \rightarrow R/I \times R/J, \quad r \mapsto (r + I, r + J).$$

Its kernel consists of elements in both I and J , so $\ker \phi = I \cap J$. To show surjectivity, choose $a \in I, b \in J$ with $a + b = 1$. Given $(x + I, y + J)$, set

$$r = xb + ya.$$

Modulo I , $a \equiv 0$ and $b = 1 - a \equiv 1$, so $r \equiv x$. Modulo J , $b \equiv 0$ and $a \equiv 1$, so $r \equiv y$. Thus ϕ is surjective, and the first isomorphism theorem gives the first conclusion.

Clearly $IJ \subseteq I \cap J$. Conversely, if $z \in I \cap J$, then using $1 = a + b$,

$$z = z(a + b) = za + zb.$$

Since $z \in J$ and $a \in I$, $za \in IJ$; similarly $zb \in IJ$. Hence $z \in IJ$. □

Intuition

The CRT says independent congruence conditions can be solved simultaneously when their moduli do not interfere. Algebraically, comaximal ideals behave like independent coordinate directions.

3.10 Definition recall drill

Without notes, state and explain:

1. group, subgroup, normal subgroup;
2. kernel and image of a homomorphism;
3. group action, orbit, stabilizer;
4. ideal, prime ideal, maximal ideal;
5. comaximal ideals;
6. the first isomorphism theorem and the CRT.

For each definition, produce one example and one nonexample or near-miss.

3.11 Exercises

Exercise: A1: subgroup test

Show that $n\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$ for every integer $n \geq 0$. Use the one-step subgroup test.

Exercise: A2: index two implies normal

Prove that every subgroup of index 2 is normal.

Exercise: A3: first isomorphism theorem in action

Consider $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}, k \mapsto k \bmod n$. Find its kernel and image, and recover $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$ from the first isomorphism theorem in a way that makes clear what information is being forgotten.

Exercise: A4: conjugacy in S_3

Determine all conjugacy classes of S_3 and verify that their sizes sum to 6.

Exercise: A5: orbit–stabilizer

Let S_4 act on the set of 2-element subsets of $\{1, 2, 3, 4\}$. Find the orbit and stabilizer of $\{1, 2\}$ and verify orbit–stabilizer.

Exercise: A6: Sylow

Let $|G| = 15$. Show that the Sylow 5-subgroup is normal. Determine the possible numbers of Sylow 3-subgroups.

Exercise: A7: ideals of $\mathbb{Z}$

Prove every ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$ for a unique $n \geq 0$.

Exercise: A8: quotient computation

Describe $\mathbb{Z}[x]/(x, 6)$ up to canonical isomorphism. Explain each simplification.

Exercise: A9: prime and maximal in $\mathbb{Z}$

For $n \geq 2$, determine when (n) is prime and when it is maximal in $\mathbb{Z}$.

Exercise: A10: CRT computation

Solve

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 4 \pmod{7}$$

and explain why the solution is unique modulo 105.

Exercise: A11: quotient intuition

Let $R = \mathbb{R}[x]$ and $I = (x^2 + 1)$. Show that every class in R/I has a representative $a + bx$. Explain why this quotient behaves like adjoining a square root of -1 .

Exercise: A12: correspondence of ideals

Let $\pi : R \rightarrow R/I$ be the quotient map. Show that ideals of R/I correspond bijectively to ideals J of R containing I , via $J \mapsto J/I$ and inverse image under π .

Mastery gate: Algebra gate

You are ready to leave this chapter when quotient groups and quotient rings feel like the same structural idea; you can explain why normal subgroups and ideals are the exact hypotheses needed for quotients; and you can derive orbit–stabilizer, the first isomorphism theorem, and the prime/maximal quotient criteria without relying on memorized wording.

Chapter 4

Algebra reboot II: fields, polynomials, and Galois theory

Galois theory is central to Chebotarev because Chebotarev describes how prime ideals distribute according to elements of a Galois group. This chapter is still a reboot, not the full Phase 2 course. The aim is to recover the meanings behind the standard definitions so that the later systematic treatment has something stable to attach to.

4.1 Fields and field extensions

Definition: field

A *field* is a commutative ring K with $1 \neq 0$ in which every nonzero element has a multiplicative inverse.

Examples are $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, finite fields such as $\mathbb{F}_p$, and rational-function fields $K(t)$. The ring $\mathbb{Z}$ is not a field because, for example, $2^{-1} \notin \mathbb{Z}$.

Intuition

A field is an arithmetic universe in which linear algebra works cleanly: nonzero scalars may be divided by. This is why a larger field L containing K can be viewed as a vector space over K .

Definition: field extension and degree

If $K \subseteq L$ are fields, L/K is a *field extension*. Since L is a vector space over K , its dimension

$$[L : K] := \dim_K L$$

is called the *degree* of the extension.

Worked example: A quadratic extension

Every element of $\mathbb{Q}(\sqrt{2})$ can be written uniquely as $a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$. Thus $\{1, \sqrt{2}\}$ is a basis over $\mathbb{Q}$ and

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2.$$

Uniqueness follows because $a + b\sqrt{2} = 0$ with $b \neq 0$ would imply $\sqrt{2} = -a/b \in \mathbb{Q}$, impossible.

4.1.1 The tower law

Theorem 4.1 (Tower law). *If $K \subseteq E \subseteq L$ and both $[E : K]$ and $[L : E]$ are finite, then*

$$[L : K] = [L : E][E : K].$$

Proof. Let $e_1, \dots, e_m$ be a K -basis of E , and $\ell_1, \dots, \ell_n$ an E -basis of L . We claim that the mn products $e_i \ell_j$ form a K -basis of L .

For spanning, any $x \in L$ has $x = \sum_j a_j \ell_j$ with $a_j \in E$. Each $a_j = \sum_i c_{ij} e_i$ with $c_{ij} \in K$, hence $x = \sum_{i,j} c_{ij} e_i \ell_j$.

For independence, suppose $\sum_{i,j} c_{ij} e_i \ell_j = 0$. Grouping by j gives

$$\sum_j \left(\sum_i c_{ij} e_i \right) \ell_j = 0.$$

Because the ℓ_j are E -linearly independent, every coefficient $\sum_i c_{ij} e_i$ is zero. Since the e_i are K -linearly independent, all $c_{ij} = 0$. Thus the products form a basis. $\square$

Memory hook

Degrees multiply because bases multiply: a basis upstairs times a basis downstairs gives a basis for the whole tower.

4.2 Algebraic elements and minimal polynomials

Definition: algebraic element

Let L/K be a field extension. An element $\alpha \in L$ is *algebraic over K* if some nonzero polynomial $f \in K[x]$ satisfies $f(\alpha) = 0$. Otherwise α is *transcendental over K* .

Definition: minimal polynomial

If α is algebraic over K , its *minimal polynomial* $m_{\alpha,K}(x)$ is the unique monic polynomial of smallest positive degree in $K[x]$ satisfying $m_{\alpha,K}(\alpha) = 0$.

Why is it irreducible? If $m = fg$ with both f, g of smaller positive degree, then

$$0 = m(\alpha) = f(\alpha)g(\alpha).$$

Because L is a field, it has no zero divisors, so one factor vanishes at α , contradicting minimality.

Conversely, if $f \in K[x]$ is monic irreducible and $f(\alpha) = 0$, then $f = m_{\alpha,K}$: divide f by m to obtain $f = qm + r$ with $\deg r < \deg m$. Evaluating at α gives $r(\alpha) = 0$, so minimality forces $r = 0$, hence $m \mid f$; irreducibility and monicity force equality.

4.2.1 The quotient construction for adjoining one algebraic element

Define evaluation

$$\text{ev}_\alpha : K[x] \rightarrow L, \quad f \mapsto f(\alpha).$$

Its image is $K[\alpha]$ and its kernel is exactly (m_α) . Therefore

$$K[x]/(m_\alpha) \cong K[\alpha].$$

Since m_α is irreducible, (m_α) is maximal in the PID $K[x]$, so the quotient is a field. Hence $K[\alpha] = K(\alpha)$. Every class modulo m_α has a unique representative of degree less than $d = \deg m_\alpha$, so

$$1, \alpha, \dots, \alpha^{d-1}$$

is a K -basis of $K(\alpha)$. In particular,

$$[K(\alpha) : K] = \deg m_\alpha.$$

Intuition: What adjoining an algebraic element means

The quotient $K[x]/(m)$ turns the formal variable x into an element satisfying the relation $m(x) = 0$. Adjoining α is therefore “ordinary polynomial arithmetic, plus the permission to simplify using the equation satisfied by α .”

Worked example: The minimal polynomial of $\sqrt{2} + \sqrt{3}$

Let $\alpha = \sqrt{2} + \sqrt{3}$. Squaring,

$$\alpha^2 = 5 + 2\sqrt{6},$$

so $\alpha^2 - 5 = 2\sqrt{6}$. Squaring again,

$$(\alpha^2 - 5)^2 = 24,$$

which gives

$$\alpha^4 - 10\alpha^2 + 1 = 0.$$

Thus α is a root of $f(x) = x^4 - 10x^2 + 1$.

To show this is minimal, observe that

$$\frac{1}{\alpha} = \sqrt{3} - \sqrt{2}$$

because $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 1$. Hence

$$\sqrt{3} = \frac{1}{2}(\alpha + \alpha^{-1}), \quad \sqrt{2} = \frac{1}{2}(\alpha - \alpha^{-1}),$$

so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Now $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$: if $\sqrt{3} = a + b\sqrt{2}$, squaring yields $3 = a^2 + 2b^2 + 2ab\sqrt{2}$, so $ab = 0$, and each case leads to a rational square equaling 3 or $3/2$, impossible. Therefore

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

The minimal polynomial of α must therefore have degree 4, so it is exactly $x^4 - 10x^2 + 1$.

4.3 Splitting fields

Definition: splitting field

For $f \in K[x]$, a *splitting field* of f over K is an extension L/K such that f factors into linear factors in $L[x]$ and L is generated over K by the roots of f .

Intuition

A splitting field is the smallest field in which the polynomial has revealed all of its roots. Galois theory studies the symmetries of this complete root configuration.

4.4 Separable and normal extensions

Definition: separable polynomial

A polynomial is *separable* if it has no repeated roots in a splitting field. An algebraic extension L/K is separable if every element of L is separable over K .

A useful derivative criterion is

$$f \text{ has a repeated root} \iff \gcd(f, f') \neq 1.$$

Indeed, if α is repeated then $(x - \alpha)^2 \mid f$, so $x - \alpha$ divides both f and f' . Conversely, a common root of f and f' has multiplicity at least 2.

Over a field of characteristic 0, every irreducible polynomial is separable. If an irreducible f had $f' = 0$, then all exponents would be multiples of the characteristic; in characteristic 0 this forces f constant, impossible. Since f is irreducible, $\gcd(f, f')$ is then 1.

Definition: normal extension

An algebraic extension L/K is *normal* if every irreducible polynomial $f \in K[x]$ having one root in L splits completely over L .

Intuition

Separability says the conjugate roots are distinct. Normality says that if the extension contains one conjugate root, it contains all of them. A finite Galois extension is precisely one where the root structure is both complete and nondegenerate.

Definition: Galois extension

A finite extension L/K is *Galois* if it is normal and separable. Its Galois group is

$$\text{Gal}(L/K) = \{\sigma : L \rightarrow L : \sigma \text{ is a field automorphism and } \sigma|_K = \text{id}\}.$$

4.5 Why automorphisms permute roots

Let $\sigma \in \text{Gal}(L/K)$ and $f(x) = \sum a_i x^i \in K[x]$. If $f(\alpha) = 0$, then

$$0 = \sigma(f(\alpha)) = \sigma\left(\sum a_i \alpha^i\right) = \sum \sigma(a_i) \sigma(\alpha)^i = \sum a_i \sigma(\alpha)^i = f(\sigma(\alpha)),$$

because σ fixes K . Therefore every K -automorphism sends a root of a polynomial over K to another root of the same polynomial.

Memory hook

A Galois automorphism cannot change equations whose coefficients lie in the base field. It may only permute their solutions.

This gives a concrete way to think about a Galois group: it is a group of permutations of roots constrained by all algebraic relations among them.

4.6 Finite fields and Frobenius

Let K have characteristic $p > 0$. The binomial theorem simplifies because

$$\binom{p}{k} \equiv 0 \pmod{p} \quad (0 < k < p).$$

Hence

$$(a + b)^p = a^p + b^p,$$

and clearly $(ab)^p = a^p b^p$. Therefore

$$F_p : x \mapsto x^p$$

is a field homomorphism, called the *Frobenius homomorphism*.

In a finite field it is automatically injective, hence surjective, and therefore an automorphism.

Theorem 4.2 (Finite-field Frobenius). *For $q = p^r$, the map*

$$F : x \mapsto x^q$$

is an automorphism of $\mathbb{F}_{q^n}$ fixing $\mathbb{F}_q$, and $F^n = \text{id}$.

Proof. Since $x \mapsto x^p$ is an automorphism, so is its r -fold iterate $x \mapsto x^{p^r} = x^q$. Every $a \in \mathbb{F}_q$ satisfies $a^q = a$: nonzero a lies in the multiplicative group $\mathbb{F}_q^\times$ of order $q - 1$, so $a^{q-1} = 1$, while $0^q = 0$. Thus F fixes $\mathbb{F}_q$.

For $x \in \mathbb{F}_{q^n}$,

$$F^n(x) = x^{q^n} = x,$$

by the same reasoning in the field with q^n elements. Therefore $F^n = \text{id}$. $\square$

In fact $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ is cyclic of order n , generated by F . A full proof belongs in Phase 2, but the essential picture should already be familiar: finite-field Galois theory comes with a canonical generator given by raising to the size of the base field.

Intuition: Why Frobenius matters for Chebotarev

Later, for an unramified prime $\mathfrak{p}$, a global Galois automorphism is characterized by inducing the finite-field map $x \mapsto x^{N\mathfrak{p}}$ on the residue field. The finite-field Frobenius in this chapter is therefore the local prototype of the Frobenius elements distributed by Chebotarev.

4.7 The fundamental theorem of Galois theory

Theorem 4.3 (Fundamental theorem, finite case). *Let L/K be a finite Galois extension with group $G = \text{Gal}(L/K)$. There is an inclusion-reversing bijection*

$$\{\text{intermediate fields } K \subseteq E \subseteq L\} \longleftrightarrow \{\text{subgroups } H \leq G\}$$

given by

$$E \mapsto \text{Gal}(L/E), \quad H \mapsto L^H := \{x \in L : \sigma(x) = x \forall \sigma \in H\}.$$

Moreover,

$$[L : E] = |\text{Gal}(L/E)|, \quad [E : K] = [G : \text{Gal}(L/E)].$$

Finally, E/K is Galois if and only if $\text{Gal}(L/E) \triangleleft G$, and then

$$\text{Gal}(E/K) \cong G / \text{Gal}(L/E).$$

A full proof uses several results that Phase 2 will revisit. For Phase 0, focus on the architecture.

Intuition: Why the correspondence reverses inclusion

A larger subgroup imposes more equations $\sigma(x) = x$, so fewer elements survive as fixed points; hence a larger subgroup corresponds to a smaller field. Conversely, a larger intermediate field gives automorphisms more elements they must fix, so fewer automorphisms remain.

The normal-subgroup clause should feel familiar: a quotient group G/H exists precisely when H is normal. If E/K is itself Galois, restricting automorphisms from L to E gives a homomorphism

$$G \rightarrow \text{Gal}(E/K)$$

whose kernel is $\text{Gal}(L/E)$; the first isomorphism theorem then produces the quotient.

4.8 Worked example: a cubic splitting field

Let $\alpha = \sqrt[3]{2}$ and $\zeta = e^{2\pi i/3}$. The roots are

$$\alpha, \quad \zeta\alpha, \quad \zeta^2\alpha.$$

Thus the splitting field is

$$L = \mathbb{Q}(\alpha, \zeta).$$

The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$ but $\zeta \notin \mathbb{R}$, we have $\zeta \notin \mathbb{Q}(\alpha)$. As ζ satisfies $x^2 + x + 1$, adjoining it has degree 2. Hence

$$[L : \mathbb{Q}] = 6.$$

Because L is the splitting field of a separable polynomial in characteristic 0, $L/\mathbb{Q}$ is Galois, so its Galois group has order 6.

Every automorphism permutes the three roots, giving an injection

$$\text{Gal}(L/\mathbb{Q}) \hookrightarrow S_3.$$

Both groups have order 6, so

$$\text{Gal}(L/\mathbb{Q}) \cong S_3.$$

Complex conjugation fixes the real element α and swaps $\zeta\alpha$ with $\zeta^2\alpha$; it corresponds to a transposition. Its fixed field has degree

$$[S_3 : \langle c \rangle] = 3$$

over $\mathbb{Q}$, and $\mathbb{Q}(\alpha)$ is visibly fixed by conjugation and has degree 3, hence

$$L^{\langle c \rangle} = \mathbb{Q}(\sqrt[3]{2}).$$

4.9 Definition recall drill

Without notes, state and explain:

1. field extension and extension degree;
2. algebraic element and minimal polynomial;
3. splitting field;
4. separable, normal, and Galois;

5. Galois group and fixed field;
6. finite-field Frobenius.

Then explain in one paragraph why “Galois automorphisms permute roots” is almost forced by the definition.

4.10 Exercises

Exercise: G1: degree by minimal polynomial

Show that $[\mathbb{Q}(\sqrt{5}) : \mathbb{Q}] = 2$ and write the inverse of $3 + \sqrt{5}$ in the basis $\{1, \sqrt{5}\}$.

Exercise: G2: tower law

Compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ and justify every step.

Exercise: G3: minimal polynomial

Find the minimal polynomial over $\mathbb{Q}$ of $\sqrt{2} + \sqrt{3}$ and prove it is minimal, not merely an annihilating polynomial.

Exercise: G4: quotient construction

Show explicitly that

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}).$$

Your solution must define the map, find its kernel, and justify surjectivity.

Exercise: G5: separability criterion

Show that $f(x) = x^4 - 2$ is separable over $\mathbb{Q}$. Then show that $g(x) = x^p - t$ in $\mathbb{F}_p(t)[x]$ has derivative zero and explain why this warns us that characteristic p behaves differently.

Exercise: G6: splitting field

Find the splitting field over $\mathbb{Q}$ of $x^4 - 2$ and give an upper bound for its degree. You do not need to determine the full Galois group.

Exercise: G7: roots move to roots

Let L/K be an extension and $\sigma \in \text{Aut}(L/K)$. Prove from first principles that if α is algebraic over K , then $\sigma(\alpha)$ has the same minimal polynomial over K as α .

Exercise: G8: finite-field Frobenius

Prove directly that $x \mapsto x^p$ is a ring homomorphism in characteristic p . Identify exactly where the characteristic assumption is used.

Exercise: G9: $\mathbb{F}_{p^n}$ roots

Show that every element of $\mathbb{F}_{p^n}$ is a root of $x^{p^n} - x$. Why does this imply that the polynomial has exactly p^n distinct roots in $\mathbb{F}_{p^n}$?

Exercise: G10: fixed field intuition

Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Define automorphisms by independently changing the signs of $\sqrt{2}$ and $\sqrt{3}$. Show that these give four automorphisms and identify the fixed field of the subgroup generated by the automorphism $\sqrt{2} \mapsto -\sqrt{2}, \sqrt{3} \mapsto \sqrt{3}$.

Exercise: G11: cyclotomic warm-up

Let $\zeta_5 = e^{2\pi i/5}$. Show that any $\mathbb{Q}$ -automorphism of $\mathbb{Q}(\zeta_5)$ sends ζ_5 to another primitive fifth root of unity. Deduce that there are at most four such automorphisms.

Exercise: G12: $x^3 - 2$

Reconstruct the calculation that the splitting field of $x^3 - 2$ has Galois group S_3 . Do so without referring to the worked example until you have a complete attempt.

Mastery gate: Field/Galois gate

Move on when you can recover extension degrees from minimal polynomials and towers; explain separable versus normal in your own words; compute the splitting field and Galois group of $x^3 - 2$; and prove the finite-field Frobenius facts without looking them up. It is acceptable if the full proof of the fundamental theorem is not yet reconstructible.

Chapter 5

Analysis reboot I: limits, series, and uniform convergence

The analytic machinery used later in the Chebotarev course depends much less on clever epsilon arguments than on knowing *which kind of convergence is strong enough to justify which operation*. This chapter therefore emphasizes quantifiers, estimates, and the logic behind exchanging limits with sums, integrals, and derivatives.

5.1 Sequences and the meaning of a limit

Definition: convergence of a sequence

A sequence (a_n) of real or complex numbers converges to L if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - L| < \varepsilon.$$

We write $a_n \rightarrow L$.

Intuition

The definition says that every demanded accuracy can eventually be met permanently. The index N may depend on the demanded tolerance ε , but once N is chosen, *all* later terms must stay inside the tolerance.

The order of quantifiers matters. “For every n there exists an ε ” would say almost nothing. A useful way to remember the correct order is: *the challenger chooses the accuracy; you respond with a sufficiently late starting point*.

Worked example: A basic epsilon proof

Show $1/n \rightarrow 0$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then for $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| \leq \frac{1}{N} < \varepsilon.$$

The mathematics is the inequality $n \geq N \Rightarrow 1/n \leq 1/N$; the quantifiers merely record who chooses what and in what order.

5.2 Cauchy sequences and completeness

Definition: Cauchy sequence

A sequence (a_n) is Cauchy if

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N : |a_n - a_m| < \varepsilon.$$

Intuition

Convergence compares late terms to a proposed external limit. The Cauchy condition compares late terms only with one another. It detects whether the sequence is trying to settle down even before the limiting object is known.

Every convergent sequence is Cauchy by the triangle inequality. In $\mathbb{R}$ and $\mathbb{C}$, the converse is true: these fields are *complete*. Completeness is the principle that limit processes do not force us to leave the space.

5.3 Infinite series

Definition: series

The series $\sum_{n=1}^{\infty} a_n$ converges if the sequence of partial sums

$$s_N = \sum_{n=1}^N a_n$$

converges.

Thus a series is not a mysterious infinite addition; it is an ordinary sequence of finite sums followed by a limit.

A necessary condition is $a_n \rightarrow 0$, because $a_n = s_n - s_{n-1}$ and convergent sequences have convergent differences. The converse is false: the harmonic series diverges.

5.3.1 Absolute and conditional convergence

Definition: absolute convergence

A series $\sum a_n$ converges *absolutely* if $\sum |a_n|$ converges. It converges *conditionally* if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Absolute convergence is stronger because it gives robust control over rearrangements and products. In analytic number theory, absolute convergence is especially valuable because it legitimizes manipulations of Dirichlet series and Euler products.

Proposition 5.1. *Absolute convergence implies convergence.*

Proof. If $\sum |a_n|$ converges, its tails become arbitrarily small. For $m > n$,

$$\left| \sum_{k=n+1}^m a_k \right| \leq \sum_{k=n+1}^m |a_k|.$$

Hence the partial sums of $\sum a_n$ are Cauchy, and completeness gives convergence. $\square$

5.3.2 Standard tests: what they are really measuring

The comparison test asks whether terms are dominated by a known summable scale. The ratio and root tests measure approximate exponential growth. The integral test compares a positive monotone sequence with area under a curve. The alternating-series test exploits cancellation.

For later work, retain especially:

- $\sum n^{-p}$ converges iff $p > 1$;
- $\sum r^n$ converges iff $|r| < 1$;
- logarithmic refinements such as $\sum 1/[n(\log n)^p]$, which converge iff $p > 1$.

5.4 Sequences of functions

Let $f_n : X \rightarrow \mathbb{R}$ or $\mathbb{C}$.

Definition: pointwise convergence

We say $f_n \rightarrow f$ pointwise on X if

$$\forall x \in X \forall \varepsilon > 0 \exists N = N(x, \varepsilon) \forall n \geq N : |f_n(x) - f(x)| < \varepsilon.$$

Definition: uniform convergence

We say $f_n \rightarrow f$ uniformly on X if

$$\forall \varepsilon > 0 \exists N = N(\varepsilon) \forall x \in X \forall n \geq N : |f_n(x) - f(x)| < \varepsilon.$$

Equivalently,

$$\sup_{x \in X} |f_n(x) - f(x)| \rightarrow 0$$

when the supremum is defined in $[0, \infty]$.

Intuition: The decisive quantifier swap

Pointwise convergence lets each point choose its own starting time N . Uniform convergence requires one starting time that works everywhere simultaneously. Uniform convergence therefore controls the entire graph at once.

Worked example: Powers on the unit interval

For $0 \leq x < 1$, $x^n \rightarrow 0$, while $1^n = 1$. Thus the pointwise limit is

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

The convergence cannot be uniform because every f_n is continuous whereas f is discontinuous, and uniform limits of continuous functions are continuous. Directly, for any n choose $x_n = 2^{-1/n} < 1$. Then

$$|f_n(x_n) - f(x_n)| = x_n^n = \frac{1}{2},$$

so the supremum error never becomes smaller than $1/2$.

5.5 Why uniform limits preserve continuity

Theorem 5.2. *If each $f_n : X \rightarrow \mathbb{R}$ is continuous at x_0 and $f_n \rightarrow f$ uniformly, then f is continuous at x_0 .*

Proof. Given $\varepsilon > 0$, choose N so that

$$|f_N(x) - f(x)| < \varepsilon/3$$

for every $x \in X$. Because f_N is continuous at x_0 , choose $\delta > 0$ so that $d(x, x_0) < \delta$ implies

$$|f_N(x) - f_N(x_0)| < \varepsilon/3.$$

Then

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \varepsilon. \end{aligned}$$

The proof is a three-term triangle-inequality bridge: limit function $\rightarrow$ one approximating function $\rightarrow$ nearby value of that approximating function $\rightarrow$ limit function. $\square$

Memory hook

Uniform convergence lets you choose *one* approximating function f_N that is good at both x and x_0 . That is why continuity can be transferred through the limit.

5.6 Uniform convergence and integration

Theorem 5.3. *If $f_n : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable and $f_n \rightarrow f$ uniformly, then f is integrable (under the usual hypotheses; in particular this follows if each f_n is continuous) and*

$$\int_a^b f_n \rightarrow \int_a^b f.$$

The essential estimate is

$$\left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n - f| \leq (b - a) \sup_{x \in [a, b]} |f_n(x) - f(x)|.$$

The right side tends to 0 by uniform convergence.

Intuition

Integration averages error over an interval. Uniform convergence gives a global ceiling on the error; multiplying that ceiling by the interval length bounds the total accumulated error.

5.7 Function series and the Weierstrass M-test

Theorem 5.4 (Weierstrass M-test). *Suppose $|f_n(x)| \leq M_n$ for every $x \in X$ and $\sum M_n < \infty$. Then*

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly and absolutely on X .

Proof. Let $S_N = \sum_{n=1}^N f_n$. For $m > n$,

$$\sup_{x \in X} |S_m(x) - S_n(x)| \leq \sum_{k=n+1}^m M_k.$$

Since $\sum M_k$ converges, its tails tend uniformly to zero. Thus (S_N) is uniformly Cauchy and converges uniformly (pointwise completeness of $\mathbb{R}$ or $\mathbb{C}$ supplies the limit). Absolute convergence follows from $\sum |f_n(x)| \leq \sum M_n$ for each x . $\square$

This theorem will later justify termwise integration and holomorphicity of series on compact subsets.

5.8 Differentiation is more delicate

Uniform convergence of f_n alone does *not* imply convergence of derivatives. A standard counterexample is

$$f_n(x) = \frac{\sin(nx)}{n},$$

which converges uniformly to 0, while

$$f'_n(x) = \cos(nx)$$

does not converge pointwise for most x .

A useful theorem is: if f'_n converges uniformly and $f_n(x_0)$ converges at one point, then under standard interval hypotheses f_n converges uniformly to a differentiable f and $f' = \lim f'_n$. The point is that differentiation amplifies small-scale oscillation, so stronger hypotheses are needed.

5.9 Power series

A power series centered at z_0 is

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

Its convergence is governed by a radius $R \in [0, \infty]$: it converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$. Boundary points require separate analysis.

Theorem 5.5 (Cauchy–Hadamard formula). *The radius is determined by*

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n},$$

with the usual conventions for 0 and ∞ .

5.9.1 Why this formula appears

Apply the root test to the terms $a_n(z - z_0)^n$:

$$\limsup |a_n(z - z_0)^n|^{1/n} = |z - z_0| \limsup |a_n|^{1/n}.$$

The root test gives convergence when this quantity is < 1 and divergence when it is > 1 . The threshold is therefore precisely $|z - z_0| = R$.

Intuition

The coefficients have an asymptotic exponential scale $|a_n|^{1/n}$. The radius is the reciprocal scale needed for $(z - z_0)^n$ to cancel that growth.

Inside the disk, convergence is much stronger than merely pointwise. If $0 < r < R$, then the series converges uniformly on the closed disk $|z - z_0| \leq r$ by the M-test: choose ρ with $r < \rho < R$. Since $\sum |a_n| \rho^n$ converges, the bounds $|a_n(z - z_0)^n| \leq |a_n| r^n$ are summable.

This local uniform convergence is one reason power series are so well behaved: they may be differentiated and integrated term by term inside their disk of convergence.

Worked example: Radius and boundary

For

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n,$$

we have

$$|a_n|^{1/n} = \frac{n^{2/n}}{3} \rightarrow \frac{1}{3},$$

so $R = 3$. On $|z| = 3$, the term has magnitude n^2 , hence does not even tend to zero; the series diverges at every boundary point.

5.10 Definition recall drill

Without notes, write:

1. sequence convergence and its negation;
2. Cauchy sequence;
3. absolute versus conditional convergence;
4. pointwise versus uniform convergence, with quantifiers in the correct order;
5. the M-test;
6. the meaning of the radius of convergence.

Then explain why uniform convergence, not pointwise convergence, is naturally suited to passing continuity and integration through a limit.

5.11 Exercises**Exercise: R1: epsilon proof**

Prove from the definition that $\frac{2n+1}{n+3} \rightarrow 2$.

Exercise: R2: Cauchy implication

Prove every convergent sequence in $\mathbb{R}$ is Cauchy.

Exercise: R3: series classification

Classify

$$\sum \frac{(-1)^{n+1}}{n}, \quad \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}, \quad \sum_{n=1}^{\infty} \frac{n!}{n^n}$$

as absolutely convergent, conditionally convergent, or divergent.

Exercise: R4: geometric tailFor $|r| < 1$, derive

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

from the finite geometric sum, and give an explicit bound on the tail after N terms.**Exercise: R5: pointwise versus uniform**Analyze $f_n(x) = x^n$ on $[0, 1]$ and on $[0, a]$ for fixed $0 < a < 1$. State exactly why the two domains behave differently.**Exercise: R6: uniform convergence by supremum**Let $f_n(x) = x/(n+x)$ on $[0, 1]$. Prove $f_n \rightarrow 0$ uniformly and find a useful bound for the sup norm.**Exercise: R7: M-test**

Show that

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

converges uniformly on $\mathbb{R}$.**Exercise: R8: interchange sum and integral**

Justify

$$\int_0^1 \sum_{n=1}^{\infty} \frac{x^n}{n^2} dx = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}.$$

Name the theorem that permits the interchange and verify its hypotheses.

Exercise: R9: uniform limit theoremReconstruct the proof that a uniform limit of continuous functions is continuous, using an $\varepsilon/3$ argument.**Exercise: R10: differentiation warning**For $f_n(x) = \sin(nx)/n$, show $f_n \rightarrow 0$ uniformly on $\mathbb{R}$ but $f'_n(0) = 1$ for every n . Explain why this disproves the naive rule “differentiate a uniformly convergent sequence termwise.”

Exercise: R11: power-series radius

Find the radius of convergence of

$$\sum_{n=1}^{\infty} n^3 z^n, \quad \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \sum_{n=1}^{\infty} \frac{z^n}{n}.$$

For the third series, analyze convergence at $z = 1$ and $z = -1$.

Exercise: R12: local uniform convergence

Let $\sum a_n z^n$ have radius $R > 0$. Prove directly, using the M-test, that it converges uniformly on every disk $|z| \leq r < R$.

Mastery gate: Real-analysis gate

Move on when you can write the quantifiers for pointwise and uniform convergence without hesitation, explain the quantifier difference intuitively, prove the uniform limit and integration results, use the M-test, and compute radii of convergence including boundary behavior.

Chapter 6

Analysis reboot II: complex analysis

Complex analysis later turns arithmetic information about primes into analytic information about zeta and L -functions. The important reboot is therefore not a catalogue of contour tricks. It is the structural chain

holomorphicity $\implies$ Cauchy theory $\implies$ rigidity $\implies$ residues and analytic continuation tools.

6.1 Complex differentiability and holomorphicity

Definition: complex derivative

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$. The derivative of f at $z_0 \in U$ is

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided the limit exists for complex h approaching 0 from *every* direction.

Definition: holomorphic

A function $f : U \rightarrow \mathbb{C}$ is *holomorphic* on U if it is complex differentiable at every point of U .

Intuition: Why complex differentiability is unusually strong

For a real derivative, h approaches zero along one line. In $\mathbb{C}$, h can approach from infinitely many directions. Requiring the same quotient limit in all directions imposes strong compatibility conditions. This is the source of the rigidity of holomorphic functions.

Write $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. If f is differentiable, approach along real increments $h \in \mathbb{R}$:

$$f'(z) = u_x + iv_x.$$

Approach along imaginary increments ih :

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z + ih) - f(z)}{ih} = v_y - iu_y.$$

Equating real and imaginary parts gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Under suitable continuity hypotheses on the first partial derivatives, these equations are also sufficient for holomorphicity.

Worked example: Why $\bar{z}$ is not holomorphic

For $f(z) = \bar{z} = x - iy$, we have $u = x$, $v = -y$. Then

$$u_x = 1, \quad v_y = -1,$$

so the Cauchy–Riemann equations fail everywhere. Equivalently, the difference quotient along real h tends to 1, while along imaginary h it tends to -1 .

6.2 Contours and complex integration

A piecewise C^1 path is a map $\gamma : [a, b] \rightarrow \mathbb{C}$. For continuous f , define

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

This is ordinary real-variable integration applied to the parametrized curve.

If γ is closed, $\gamma(a) = \gamma(b)$. Orientation matters: reversing a path changes the sign of the integral.

Intuition

The factor $\gamma'(t)$ converts movement in the parameter t into movement in the complex plane. A contour integral therefore accumulates the values of f weighted by the direction and speed of travel.

6.3 Cauchy's theorem

A standard form says: if f is holomorphic on a simply connected open set U , then

$$\int_{\gamma} f(z) dz = 0$$

for every closed piecewise C^1 curve γ in U .

The precise topological hypotheses can vary among formulations. The conceptual point is that a holomorphic function has no local obstruction to possessing an antiderivative, and therefore closed contour integrals vanish when there are no holes or singularities trapped inside.

Memory hook

Nonzero closed contour integrals are evidence of a singularity or topological obstruction inside the loop. This is the intuition behind residues.

6.4 Cauchy's integral formula

Theorem 6.1 (Cauchy integral formula). *Suppose f is holomorphic on an open set containing a positively oriented simple closed contour γ and its interior. If z_0 lies inside γ , then*

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz.$$

6.4.1 Why the formula is plausible

Write

$$\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}.$$

The first term has a removable singularity at z_0 : by differentiability,

$$\frac{f(z) - f(z_0)}{z - z_0} \rightarrow f'(z_0).$$

After filling in that value, it is holomorphic, so its integral around γ vanishes by Cauchy's theorem. Hence

$$\int_{\gamma} \frac{f(z)}{z - z_0} dz = f(z_0) \int_{\gamma} \frac{dz}{z - z_0}.$$

For a positively oriented loop winding once around z_0 ,

$$\int_{\gamma} \frac{dz}{z - z_0} = 2\pi i.$$

This gives the formula.

Intuition: The surprising content

A holomorphic function's value at one interior point is determined by all of its values on a surrounding boundary. This is far stronger than ordinary differentiability and is the source of many rigidity theorems.

Differentiating under the integral gives the higher derivative formula

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Thus a holomorphic function is automatically infinitely differentiable and locally represented by a power series.

6.5 Cauchy estimates and Liouville's theorem

For a circle $|z - z_0| = R$ and $M = \max_{|z - z_0| = R} |f(z)|$, Cauchy's derivative formula yields

$$|f^{(n)}(z_0)| \leq \frac{n!M}{R^n}.$$

Indeed,

$$\begin{aligned} |f^{(n)}(z_0)| &\leq \frac{n!}{2\pi} \int_{|z - z_0| = R} \frac{|f(z)|}{|z - z_0|^{n+1}} |dz| \\ &\leq \frac{n!}{2\pi} \frac{M}{R^{n+1}} (2\pi R) = \frac{n!M}{R^n}. \end{aligned}$$

Theorem 6.2 (Liouville). *Every bounded entire function is constant.*

Proof. Suppose $|f(z)| \leq M$ on all of $\mathbb{C}$. Apply the Cauchy estimate at any z_0 with arbitrarily large R :

$$|f'(z_0)| \leq \frac{M}{R}.$$

Letting $R \rightarrow \infty$ gives $f'(z_0) = 0$. Since z_0 was arbitrary, $f' \equiv 0$, so f is constant. $\square$

This proof illustrates a recurring analytic-number-theory strategy: obtain an estimate depending on a free scale, then send the scale to a limiting regime that forces a quantity to vanish.

6.6 Zeros and the identity theorem

If a holomorphic f is not identically zero and $f(z_0) = 0$, then its Taylor series has a first nonzero term:

$$f(z) = (z - z_0)^m g(z),$$

where $m \geq 1$ and g is holomorphic with $g(z_0) \neq 0$. The integer m is the *order* or *multiplicity* of the zero.

Thus zeros of a nonzero holomorphic function are isolated. Consequently:

Theorem 6.3 (Identity theorem). *If two holomorphic functions on a connected open set agree on a set having an accumulation point inside the domain, then they agree everywhere.*

Intuition

Holomorphic functions cannot be modified locally without affecting them globally. A set of values with an interior accumulation point already pins down the whole function.

This principle is crucial for analytic continuation: once a holomorphic extension exists, it is unique on connected overlaps.

6.7 Laurent series and isolated singularities

Around an isolated singularity z_0 , a holomorphic function on an annulus can be expanded as

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

The negative-power part is the principal part.

Definition: isolated singularities

For an isolated singularity z_0 :

- it is *removable* if the principal part is zero;
- it is a *pole of order m* if finitely many negative terms occur and the lowest is $(z - z_0)^{-m}$ with nonzero coefficient;
- it is *essential* if infinitely many negative terms occur.

A pole of order m can equivalently be characterized by

$$f(z) = \frac{g(z)}{(z - z_0)^m}, \quad g(z_0) \neq 0,$$

with g holomorphic near z_0 .

6.8 Residues

Definition: residue

If

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

near an isolated singularity z_0 , then

$$\text{Res}(f; z_0) := a_{-1}.$$

Intuition: Why only the $(z - z_0)^{-1}$ coefficient matters

Around the circle $z - z_0 = re^{it}$,

$$\int (z - z_0)^n dz$$

vanishes for every integer $n \neq -1$, while for $n = -1$ it equals $2\pi i$. Integration around a loop annihilates every Laurent mode except the -1 mode.

For a simple pole of $f = g/h$ at z_0 , where $h(z_0) = 0$ and $h'(z_0) \neq 0$,

$$\text{Res}(f; z_0) = \frac{g(z_0)}{h'(z_0)}.$$

This follows from $h(z) = (z - z_0)h_1(z)$ with $h_1(z_0) = h'(z_0)$.

Theorem 6.4 (Residue theorem). *If f is meromorphic inside and on a positively oriented simple closed contour γ , with no poles on γ and poles $a_1, \dots, a_k$ inside, then*

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^k \text{Res}(f; a_j).$$

The theorem says that a global contour integral is determined by finitely many local coefficients at singularities.

Worked example: A rational contour integral

For positively oriented $|z| = 3$,

$$f(z) = \frac{z^2 + 1}{z(z-1)(z-2)}$$

has simple poles 0, 1, 2 inside. Their residues are

$$\text{Res}(f; 0) = \frac{1}{(-1)(-2)} = \frac{1}{2},$$

$$\text{Res}(f; 1) = \frac{2}{1(1-2)} = -2,$$

$$\text{Res}(f; 2) = \frac{5}{2(2-1)} = \frac{5}{2}.$$

The sum is 1, so

$$\int_{|z|=3} f(z) dz = 2\pi i.$$

6.9 Logarithmic derivatives

If f has a zero of order m at z_0 , write

$$f(z) = (z - z_0)^m g(z), \quad g(z_0) \neq 0.$$

Then

$$\frac{f'(z)}{f(z)} = \frac{m}{z - z_0} + \frac{g'(z)}{g(z)},$$

so

$$\operatorname{Res}\left(\frac{f'}{f}; z_0\right) = m.$$

If f has a pole of order m , the residue is $-m$.

This is the local engine behind the argument principle and also explains why logarithmic derivatives of Euler products are powerful: multiplicative data become additive, and zeros/poles become simple poles carrying integer weights.

Memory hook: Later connection

For zeta and L -functions, logarithmic differentiation turns an Euler product over primes into a sum supported on prime powers. This is one of the key ways complex analysis “sees” primes.

6.10 Uniform limits of holomorphic functions

A locally uniform limit of holomorphic functions is holomorphic. More precisely, if f_n are holomorphic on an open set U and converge uniformly on every compact subset of U to f , then f is holomorphic.

One route to the theorem uses Cauchy’s integral formula: on a small circle contained in U , uniform convergence allows passage of the limit through the contour integral, yielding the Cauchy formula for f . This is why compact-uniform convergence (also called locally uniform convergence) is the natural topology for holomorphic functions.

6.11 Definition recall drill

Without notes, state and explain:

1. complex derivative and holomorphic function;
2. contour integral;
3. Cauchy’s theorem and Cauchy’s integral formula;
4. zero of order m and pole of order m ;
5. residue;
6. meromorphic function;
7. locally uniform convergence.

Then explain why complex differentiability is stronger than real differentiability and why the residue is specifically the coefficient of $(z - z_0)^{-1}$.

6.12 Exercises

Exercise: C1: Cauchy–Riemann

Determine where $f(z) = z^2 \bar{z}$ is complex differentiable. Use either the Cauchy–Riemann equations or directional difference quotients.

Exercise: C2: parametrized integral

Compute directly from a parametrization

$$\int_{|z|=R} \frac{dz}{z}$$

for positive orientation.

Exercise: C3: Cauchy formula

Evaluate

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz.$$

State all relevant hypotheses.

Exercise: C4: derivative form of Cauchy

Evaluate

$$\int_{|z|=2} \frac{e^z}{(z-1)^3} dz.$$

Exercise: C5: Cauchy estimates

Suppose f is entire and $|f(z)| \leq A + B|z|$ for constants $A, B \geq 0$. Prove that f is a polynomial of degree at most 1.

Exercise: C6: removable singularity

Show that

$$f(z) = \frac{e^z - 1}{z}$$

has a removable singularity at 0 and find the value that makes the extension holomorphic.

Exercise: C7: residues at simple poles

Find the residues of

$$\frac{1}{z(z-1)}, \quad \frac{e^z}{z^2+1}$$

at all simple poles.

Exercise: C8: residue theorem

Compute

$$\int_{|z|=3} \frac{z^2+1}{z(z-1)(z-2)} dz.$$

Exercise: C9: higher-order pole

Compute

$$\operatorname{Res}\left(\frac{e^z}{(z-1)^3}; 1\right).$$

Exercise: C10: logarithmic derivative

If f is holomorphic near a and has a zero of exact order m there, prove

$$\operatorname{Res}(f'/f; a) = m.$$

Then state the analogous result for a pole of order m .

Exercise: C11: identity theorem

Suppose f and g are holomorphic on a connected open set U and agree on a sequence of distinct points $z_n \rightarrow z_0 \in U$. Prove $f \equiv g$ on U .

Exercise: C12: locally uniform series

Show that

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

converges uniformly on every closed disk $|z| \leq R$. Explain why the resulting function is entire.

Mastery gate: Complex-analysis gate

Move on when you can explain why holomorphicity is rigid; use Cauchy's formula and its derivative form; classify isolated poles in routine examples; compute residues; and explain the role of the logarithmic derivative. You do not need to reconstruct a full proof of Cauchy's theorem from Green's theorem or homotopy theory at this stage.

Chapter 7

Synthesis: reconnecting the subjects

The purpose of Phase 0 is not to leave you with five disconnected refreshers. Chebotarev lives precisely at the interfaces. This chapter rehearses several interfaces before the full theory begins.

7.1 Roots of unity: a first algebraic bridge

For $n \geq 1$, the n th roots of unity are

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}.$$

They form a cyclic group under multiplication. If $\zeta_n = e^{2\pi i/n}$, then

$$\mu_n = \{1, \zeta_n, \dots, \zeta_n^{n-1}\}.$$

The primitive roots are those of exact order n , namely ζ_n^a with $\gcd(a, n) = 1$.

This one object simultaneously involves complex analysis/geometry (points on the unit circle), finite abelian groups, polynomial roots, and field extensions. Cyclotomic fields $\mathbb{Q}(\zeta_n)$ will later connect congruence classes of primes with Galois automorphisms.

7.2 Finite-field factorization and Frobenius

Let $f \in \mathbb{F}_q[x]$ be irreducible of degree d , and let α be a root in a splitting field. The Frobenius map $F(x) = x^q$ sends roots to roots because coefficients lie in $\mathbb{F}_q$. Thus

$$\alpha, \alpha^q, \alpha^{q^2}, \dots$$

are conjugate roots. The first time the orbit returns to α is related to the degree d .

This is a prototype for a later principle: factorization behavior of a polynomial modulo a prime is encoded by the cycle structure of a Frobenius automorphism.

Intuition

Think of Frobenius as “walking around the conjugate roots.” The lengths of its cycles record the degrees of irreducible factors over the finite base field.

7.3 From unique factorization to an Euler product

For real $s > 1$, the Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Unique factorization suggests

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Here is the finite-stage derivation. For a finite set of primes P ,

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \prod_{p \in P} (1 + p^{-s} + p^{-2s} + \dots).$$

Expanding the product, each choice of exponent $e_p \geq 0$ contributes

$$\prod_{p \in P} p^{-e_p s} = n^{-s},$$

where n has no prime factors outside P . Unique factorization says every such n occurs exactly once. Therefore

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ \text{prime factors of } n \text{ lie in } P}} n^{-s}.$$

As P increases through all primes, absolute convergence for $s > 1$ justifies the limit and yields the Euler product.

Intuition: Arithmetic becomes analysis

Prime factorization is multiplicative and discrete. The Euler product packages it into a complex/real analytic function. Later we will study that analytic function near special points and extract statements about the distribution of primes.

7.4 Why conjugacy classes appear naturally

Suppose a group G acts transitively on a collection of choices. An element attached to one choice often transforms by conjugation when the choice changes. Schematically, if changing the reference point by g replaces σ by

$$g\sigma g^{-1},$$

then the individual element is not canonical, but its conjugacy class is.

This exact pattern will occur for Frobenius elements: choosing a different prime $\mathfrak{p}$ above the same base prime conjugates the Frobenius. The conjugacy class survives the choice and therefore is the natural invariant.

7.5 The emerging Chebotarev chain

You are not expected to know the objects in the middle yet, but the eventual architecture is worth seeing:

$$\text{prime} \rightarrow \text{prime ideal} \rightarrow \text{residue field} \rightarrow \text{Frobenius} \rightarrow \text{conjugacy class} \rightarrow L\text{-function} \rightarrow \text{density}$$

Phase 0 has refreshed the languages that appear at the two ends: algebra/Galois theory on the left and convergence/complex analysis on the right.

7.6 Exercises

Exercise: S1: roots of unity

Prove μ_n is a cyclic group of order n generated by $\zeta_n = e^{2\pi i/n}$. Show that ζ_n^a is a generator iff $\gcd(a, n) = 1$.

Exercise: S2: Frobenius orbit

Over $\mathbb{F}_2$, factor $x^3 + x + 1$ or prove it irreducible. If α is a root in $\mathbb{F}_8$, list the Frobenius conjugates $\alpha, \alpha^2, \alpha^4$ and explain why the next iterate returns to α .

Exercise: S3: finite Euler product

For $P = \{2, 3\}$ and $s > 1$, expand

$$(1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1}$$

as a sum and describe exactly which positive integers occur.

Exercise: S4: first Euler-product proof

Give a rigorous proof for real $s > 1$ that

$$\sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}.$$

You may use absolute convergence of $\sum n^{-s}$ and monotone convergence of positive partial sums.

Exercise: S5: automorphisms and conjugate roots

Let L/K be Galois and $f \in K[x]$. Prove that $\text{Gal}(L/K)$ acts on the set of roots of f lying in L . What are the orbits when f is irreducible and splits in L ?

Exercise: S6: why conjugacy is invariant

Suppose an element $\sigma_x \in G$ attached to a choice x satisfies $\sigma_{g \cdot x} = g\sigma_x g^{-1}$. Prove that the conjugacy class of σ_x is independent of the choice within the orbit of x .

Exercise: S7: logarithmic derivative of a finite Euler product

For finitely many primes P , define

$$F_P(s) = \prod_{p \in P} (1 - p^{-s})^{-1}.$$

Show

$$-\frac{F'_P(s)}{F_P(s)} = \sum_{p \in P} \sum_{k \geq 1} (\log p) p^{-ks}.$$

Explain in words what has happened to the multiplicative prime product.

Exercise: S8: concept map

Without notes, draw a dependency graph connecting the following terms: quotient, ideal, field extension, minimal polynomial, Galois automorphism, finite-field Frobenius, uniform convergence, holomorphic function, residue, Euler product. Every arrow must be labeled by a sentence explaining the connection.

Mastery gate: Synthesis gate

You do not need to know Chebotarev yet. You do need to see why the course can plausibly connect algebraic symmetries of roots with analytic functions built from primes. If “Euler product” and “Frobenius” still feel like facts from unrelated subjects, revisit this chapter after the solutions.

Chapter 8

A practical Phase 0 schedule and final readiness test

8.1 Ten-day default schedule

This is a default, not a deadline. If several diagnostic problems were N rather than U, double the time for the relevant days.

Day	Focus	Required output
1	Baseline diagnostic	Closed-book attempt; C/U/N calibration.
2	Proof language	Definitions from memory; P1–P8.
3	Groups and homomorphisms	Reconstruct first isomorphism theorem; A1–A6.
4	Rings, ideals, CRT	Reconstruct quotient criteria and CRT; A7–A12.
5	Fields/minimal polynomials	G1–G6; derive tower law and quotient construction.
6	Galois/Frobenius	G7–G12; reconstruct $x^3 - 2$ example.
7	Real analysis	R1–R6; memorize quantifier patterns by reconstruction.
8	Uniform convergence/power series	R7–R12; reconstruct M-test and uniform-limit proof.
9	Complex analysis	C1–C12 selectively; emphasize Cauchy formula and residues.
10	Synthesis + final test	S1–S8, then the readiness test closed-book.

8.2 How to study a definition

For every boxed definition in this workbook, use a four-card recall pattern:

1. **Name** → **formal statement**. Can I write it accurately?
2. **Name** → **intuition**. What is the definition trying to capture?
3. **Name** → **example/nonexample**. Can I generate both?
4. **Hypothesis audit**. What would fail if I removed one clause?

This is more durable than rereading highlighted definitions.

8.3 Final readiness test

Allow 2.5–3 hours. No notes. For every proof problem, write one sentence at the top identifying the decisive idea before writing details.

Exercise: F1: definitions under pressure

State accurately, and give one example of each: normal subgroup, prime ideal, minimal polynomial, Galois extension, uniform convergence, residue.

Exercise: F2: quotient structure

Prove that an ideal $\mathfrak{m}$ of a commutative ring R is maximal iff $R/\mathfrak{m}$ is a field. Then explain in one sentence why the analogous prime-ideal criterion uses “integral domain” rather than “field.”

Exercise: F3: group action and conjugacy

For the conjugation action of S_3 on itself, compute every orbit and stabilizer size. Verify orbit–stabilizer in each case.

Exercise: F4: field degree

Find the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$, prove it is minimal, and determine the degree of the generated field.

Exercise: F5: finite-field Frobenius

Prove that $x \mapsto x^q$ is an $\mathbb{F}_q$ -automorphism of $\mathbb{F}_{q^n}$ and that its n th iterate is the identity. Explain why this fact is a natural precursor to the Frobenius elements in Chebotarev.

Exercise: F6: Galois correspondence

Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Determine its Galois group and identify the intermediate quadratic fields together with the subgroups that fix them.

Exercise: F7: uniform convergence

Let $f_n(x) = x^n$ on $[0, 1]$. Determine the pointwise limit, prove convergence is not uniform, and explain exactly which quantifier prevents uniformity. Then prove convergence is uniform on every $[0, a]$ with $a < 1$.

Exercise: F8: power series

Find the radius and boundary behavior of

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n.$$

Then state why power series converge uniformly on smaller closed disks.

Exercise: F9: complex integration

Evaluate

$$\int_{|z|=2} \frac{e^z}{(z-1)^2} dz$$

and

$$\int_{|z|=3} \frac{z^2+1}{z(z-1)(z-2)} dz.$$

State the theorem used in each case.

Exercise: F10: integration of algebra and analysis

Explain, in a coherent page or less, the chain

unique factorization → Euler product → logarithmic derivative → prime-weighted sums.

Your answer should include at least one displayed formula at each transition.

8.4 Readiness rubric

Score each problem from 0 to 4:

- 4: correct, self-contained, and conceptually explained;
- 3: essentially correct with minor omissions or algebra slips;
- 2: correct central idea but substantial gaps;
- 1: relevant fragments but no complete route;
- 0: no workable approach.

Maximum score: 40.

Score	Recommendation
34–40	Move to Phase 1. Repair isolated mistakes while proceeding.
28–33	Move to Phase 1 if no single domain scored below half marks; otherwise spend 2–4 focused days repairing that domain.
20–27	Spend another week on the two weakest chapters, then retake only the corresponding test problems.
Below 20	Work through the refreshers and solutions more slowly before Phase 1. This is a retrieval issue, not a verdict on prior mathematical ability.

Mastery gate: Non-numerical override

Even with a high score, spend more time in Phase 0 if you cannot state central definitions without recognition cues. Conversely, if you score modestly because of calculation slips but can reconstruct definitions, examples, and proof architectures, it is often better to proceed and let Phase 1 provide continued practice.

Appendix A

Complete solutions

Use these solutions diagnostically. Before reading a solution to a proof problem, write down the first point at which your own attempt became uncertain. Then compare that point with the solution's decisive move. The goal is to repair the missing transition, not to memorize the prose.

A.1 Solutions to the baseline diagnostic

Solution: D1

If $n = p$ is prime, every nonzero class $\bar{a} \in \mathbb{Z}/p\mathbb{Z}$ has $p \nmid a$, hence $\gcd(a, p) = 1$. Bezout gives $ua + vp = 1$, so $\bar{u}\bar{a} = \bar{1}$; thus every nonzero class is invertible.

Conversely, if n is composite, write $n = ab$ with $1 < a, b < n$. Then $\bar{a}$ and $\bar{b}$ are nonzero in $\mathbb{Z}/n\mathbb{Z}$ but $\bar{a}\bar{b} = \bar{0}$. A field has no zero divisors, so $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Solution: D2

Write $x = 2 + 3k$. Modulo 5, $2 + 3k \equiv 3$, so $3k \equiv 1$ and hence $k \equiv 2 \pmod{5}$. Write $k = 2 + 5t$, giving $x = 8 + 15t$. Modulo 7,

$$8 + 15t \equiv 1 + t \equiv 4,$$

so $t \equiv 3 \pmod{7}$. Hence

$$x \equiv 8 + 45 = 53 \pmod{105}.$$

For pairwise comaximal ideals $I_1, \dots, I_r$,

$$R/\bigcap_i I_i \cong \prod_i R/I_i,$$

and pairwise comaximality also gives $\bigcap_i I_i = \prod_i I_i$.

Solution: D3

For $k \in \ker \varphi$ and $g \in G$,

$$\varphi(gkg^{-1}) = \varphi(g)e\varphi(g)^{-1} = e,$$

so $\ker \varphi$ is normal. Define

$$\bar{\varphi} : G/\ker \varphi \rightarrow \operatorname{im} \varphi, \quad g\ker \varphi \mapsto \varphi(g).$$

If $g\ker \varphi = h\ker \varphi$, then $h^{-1}g \in \ker \varphi$, so $\varphi(g) = \varphi(h)$; hence the map is well-defined. It is a homomorphism, surjective by definition of the image, and injective because its kernel is the identity coset. Thus it is an isomorphism.

Solution: D4

The orbit of 1 is $\{1, 2, 3\}$, so it has size 3. The stabilizer consists of the identity and the transposition (23) , so it has size 2. Thus $6 = 3 \cdot 2$, as orbit–stabilizer predicts.

Under conjugation, permutations have the same cycle type. Hence the conjugacy classes are

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

Solution: D5

For a Sylow 7-subgroup, $n_7 \mid 3$ and $n_7 \equiv 1 \pmod{7}$. The only possibility is $n_7 = 1$, so it is unique and therefore normal. For Sylow 3-subgroups, $n_3 \mid 7$ and $n_3 \equiv 1 \pmod{3}$, so $n_3 = 1$ or 7.

Solution: D6

In R/I ,

$$(a + I)(b + I) = 0 + I \iff ab \in I.$$

Thus R/I has no zero divisors exactly when $ab \in I$ forces $a \in I$ or $b \in I$, which is precisely primality. For maximality, if I is maximal and $a \notin I$, then $I + (a) = R$, so $1 = i + ra$ for some $i \in I$, and therefore $(r + I)(a + I) = 1 + I$. Thus every nonzero class is invertible. Conversely, ideals of R/I correspond to ideals of R containing I ; a field has only the ideals 0 and itself, so no proper ideal lies strictly between I and R .

Solution: D7

Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = 5 + 2\sqrt{6},$$

so $(\alpha^2 - 5)^2 = 24$, yielding

$$\alpha^4 - 10\alpha^2 + 1 = 0.$$

Also $1/\alpha = \sqrt{3} - \sqrt{2}$, hence both $\sqrt{2}$ and $\sqrt{3}$ lie in $\mathbb{Q}(\alpha)$. Therefore $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which has degree 4 over $\mathbb{Q}$. Thus the degree-4 polynomial above is minimal.

Solution: D8

In characteristic p , $x \mapsto x^p$ is a homomorphism because $(a + b)^p = a^p + b^p$. Its r th iterate is $x \mapsto x^{p^r} = x^q$, hence F is a field homomorphism. On a finite field, an injective homomorphism is surjective; injectivity follows because the kernel of a field homomorphism fixing 1 is 0. For $a \in \mathbb{F}_q$, $a^q = a$, so F fixes $\mathbb{F}_q$. Finally, every $x \in \mathbb{F}_{q^n}$ satisfies $x^{q^n} = x$, hence $F^n = \text{id}$.

Solution: D9

Let $\alpha = \sqrt[3]{2}$ and $\zeta = e^{2\pi i/3}$. The splitting field is $L = \mathbb{Q}(\alpha, \zeta)$. Eisenstein at 2 gives $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$ but $\zeta \notin \mathbb{R}$, adjoining ζ has degree 2, so $[L : \mathbb{Q}] = 6$. The Galois group embeds into S_3 by permuting the three roots; since both have order 6, it is S_3 .

Complex conjugation fixes α and has order 2. Its fixed field therefore has degree $6/2 = 3$ over $\mathbb{Q}$, and $\mathbb{Q}(\alpha)$ is a degree-3 subfield fixed by conjugation. Hence the fixed field is $\mathbb{Q}(\sqrt[3]{2})$.

Solution: D10

The alternating harmonic series converges by the alternating-series test, while its absolute series $\sum 1/n$ diverges, so it is conditionally convergent.

The series $\sum_{n \geq 2} 1/[n(\log n)^2]$ converges by the integral test: with $u = \log x$,

$$\int_2^\infty \frac{dx}{x(\log x)^2} = \int_{\log 2}^\infty u^{-2} du < \infty.$$

It is positive, hence absolutely convergent.

For $a_n = n!/n^n$,

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \left(\frac{n}{n+1}\right)^n \rightarrow e^{-1} < 1.$$

Thus the series converges absolutely by the ratio test.

Solution: D11

For $0 \leq x < 1$, $x^n \rightarrow 0$, while at $x = 1$ the value is always 1. Hence the pointwise limit is $f = 0$ on $[0, 1)$ and $f(1) = 1$.

It is not uniform. For every n , choose $x_n = 2^{-1/n} < 1$. Then $f(x_n) = 0$ but $f_n(x_n) = 1/2$, so

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| \geq \frac{1}{2}.$$

The uniform error therefore does not tend to zero.

Solution: D12

Uniform convergence means $\|f_n - f\|_\infty \rightarrow 0$. Then

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq (b-a) \|f_n - f\|_\infty \rightarrow 0.$$

Solution: D13

Here $a_n = n^2/3^n$, so

$$\limsup |a_n|^{1/n} = \frac{1}{3} \lim n^{2/n} = \frac{1}{3}.$$

Thus $R = 3$. If $|z| = 3$, the magnitude of the n th term is n^2 , which does not tend to zero. Hence the series diverges at every boundary point.

Solution: D14

Cauchy's integral formula gives

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz = e.$$

The function e^z is holomorphic on and inside the circle, 1 lies inside, and the contour is positively oriented.

Solution: D15

All poles 0, 1, 2 lie inside $|z| = 3$. The residues are $1/2, -2, 5/2$, whose sum is 1. Therefore the residue theorem gives

$$\int_{|z|=3} \frac{z^2 + 1}{z(z-1)(z-2)} dz = 2\pi i.$$

Solution: D16

Assume for contradiction that the primes are $p_1, \dots, p_r$. Set

$$N = p_1 p_2 \cdots p_r + 1.$$

Every integer $N > 1$ has a prime divisor q . But no p_i divides N , since $N \equiv 1 \pmod{p_i}$. Thus q is a prime not on the supposedly complete list, a contradiction. Contradiction enters at the initial assumption that the list of all primes is finite; the construction produces an object incompatible with that assumption.

Solution: D17

Induct on $d = \deg f$. The result is clear for $d = 0$. If f has no root, there is nothing to prove. If a is a root, polynomial division gives

$$f(x) = (x - a)g(x), \quad \deg g = d - 1.$$

Every root $b \neq a$ of f is a root of g , because $(b - a)g(b) = 0$ and $b - a \neq 0$ in a field. By induction g has at most $d - 1$ roots, so f has at most d .

Solution: D18

By the extreme value theorem, a continuous function on the compact interval $[0, 1]$ attains its minimum: there exists x_0 with

$$f(x_0) = \min_{x \in [0,1]} f(x).$$

Since $f(x) > 0$ for every x , $m := f(x_0) > 0$. Hence $f(x) \geq m$ for all x . Compactness is used exactly in the theorem that a continuous function attains its minimum; on a noncompact domain a positive continuous function may have infimum 0 without ever taking the value 0, e.g. e^{-x} on $[0, \infty)$.

A.2 Solutions to proof-language exercises**Solution: P1**

Uniform convergence means

$$\forall \varepsilon > 0 \exists N \forall n \geq N \forall x \in X : |f_n(x) - f(x)| < \varepsilon.$$

Negating one quantifier at a time gives

$$\exists \varepsilon_0 > 0 \forall N \exists n \geq N \exists x \in X : |f_n(x) - f(x)| \geq \varepsilon_0.$$

In words: there is a fixed error tolerance that fails somewhere in X arbitrarily far out in the sequence.

Solution: P2

If $x \in A \cap (B \cup C)$, then $x \in A$ and either $x \in B$ or $x \in C$. In the first case $x \in A \cap B$; in the second $x \in A \cap C$. Thus $x \in (A \cap B) \cup (A \cap C)$.

Conversely, if $x \in (A \cap B) \cup (A \cap C)$, then either $x \in A \cap B$ or $x \in A \cap C$. In either case $x \in A$ and $x \in B \cup C$, so $x \in A \cap (B \cup C)$.

Solution: P3

$12 \mid n$ implies $6 \mid n$, so P is sufficient for Q , and Q is necessary for P . The converse is false: $n = 6$ satisfies $6 \mid n$ but not $12 \mid n$.

Solution: P4

Suppose h and k are both inverses of g . Then

$$h = he = h(gk) = (hg)k = ek = k.$$

The key idea is to place both candidates in the same associative product.

Solution: P5

The statement is $n^2 \text{ odd} \Rightarrow n \text{ odd}$. Its contrapositive is $n \text{ even} \Rightarrow n^2 \text{ even}$. If $n = 2k$, then $n^2 = 4k^2 = 2(2k^2)$, so n^2 is even. Therefore the contrapositive, and hence the original statement, is true.

Solution: P6

For $n = 1$, both sides equal 1. Assume

$$1 + \dots + k = \frac{k(k+1)}{2}.$$

Then

$$1 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Thus the formula holds for $k+1$. The induction hypothesis is used in replacing the first k terms by $k(k+1)/2$.

Solution: P7

On the interval $(0, 1)$, the function $f(x) = x$ is continuous but does not attain a maximum: its supremum is 1, which is not in the domain. A correct theorem is: every continuous real-valued function on a compact set attains both its maximum and minimum. In particular, every continuous $f : [a, b] \rightarrow \mathbb{R}$ attains a maximum and a minimum.

Solution: P8

Input: Assume finitely many primes $p_1, \dots, p_r$.

Key move: Form $N = p_1 \cdots p_r + 1$; any prime divisor of N divides none of the p_i .

Output: A new prime exists, contradicting completeness of the list.

A.3 Solutions to algebra exercises

Solution: A1

The set $n\mathbb{Z}$ is nonempty since $0 \in n\mathbb{Z}$. If $a = nr$ and $b = ns$, then in the additive group the one-step subgroup condition is $a - b \in n\mathbb{Z}$:

$$a - b = n(r - s).$$

Hence $n\mathbb{Z} \leq (\mathbb{Z}, +)$.

Solution: A2

Let $H \leq G$ have index 2. There are only two left cosets: H and its complement $G \setminus H$; similarly there are only two right cosets. For $g \in H$, $gH = H = Hg$. For $g \notin H$, both gH and Hg are cosets distinct from H , hence both equal $G \setminus H$. Therefore $gH = Hg$ for every g , so $H \triangleleft G$.

Solution: A3

For $\varphi(k) = \bar{k}$, the kernel is $n\mathbb{Z}$, and the map is surjective, so the image is all of $\mathbb{Z}/n\mathbb{Z}$. The first isomorphism theorem gives

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z},$$

where the left side means the quotient group $\mathbb{Z}/(n\mathbb{Z})$. Conceptually the map forgets the exact integer and retains only its residue class; integers differing by a multiple of n become indistinguishable.

Solution: A4

Conjugation in S_3 preserves cycle type, and any two permutations of the same cycle type are conjugate. Hence the classes are

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

Their sizes are 1, 3, 2, summing to 6.

Solution: A5

The action on two-element subsets is transitive, so the orbit of $\{1, 2\}$ is the set of all $\binom{4}{2} = 6$ two-element subsets. Its stabilizer consists of permutations that preserve the set $\{1, 2\}$ as a set: independently swap 1, 2 and independently swap 3, 4. Thus it has $2 \cdot 2 = 4$ elements. Orbit-stabilizer gives $24 = 6 \cdot 4$.

Solution: A6

For $|G| = 15 = 3 \cdot 5$, the number n_5 divides 3 and satisfies $n_5 \equiv 1 \pmod{5}$, so $n_5 = 1$; the Sylow 5-subgroup is normal. The number n_3 divides 5 and satisfies $n_3 \equiv 1 \pmod{3}$, so $n_3 = 1$ or 5.

Solution: A7

Let $I \triangleleft \mathbb{Z}$. If $I = 0$, take $n = 0$. Otherwise I contains positive integers; by well-ordering let n be the least positive element of I . Since I is an ideal, $n\mathbb{Z} \subseteq I$. For any $a \in I$, divide $a = qn + r$ with $0 \leq r < n$. Then

$$r = a - qn \in I.$$

Minimality of n forces $r = 0$, so $a \in n\mathbb{Z}$. Hence $I = n\mathbb{Z}$. If $n\mathbb{Z} = m\mathbb{Z}$ with $m, n \geq 0$, each generator divides the other, so $m = n$.

Solution: A8

The ideal $(x, 6)$ forces both $x = 0$ and $6 = 0$ in the quotient. Consider evaluation at 0 followed by reduction mod 6:

$$\phi : \mathbb{Z}[x] \rightarrow \mathbb{Z}/6\mathbb{Z}, \quad f(x) \mapsto f(0) \bmod 6.$$

It is surjective. Its kernel consists of polynomials whose constant term is divisible by 6, exactly $(x, 6)$. Therefore

$$\mathbb{Z}[x]/(x, 6) \cong \mathbb{Z}/6\mathbb{Z}.$$

Solution: A9

$\mathbb{Z}/(n)$ is an integral domain iff n is prime, and it is a field iff n is prime. Therefore for $n \geq 2$, (n) is prime iff n is prime, and maximal iff n is prime. In $\mathbb{Z}$, every nonzero prime ideal is maximal.

Solution: A10

As in D2, the solution is $x \equiv 53 \pmod{105}$. Uniqueness follows because 3, 5, 7 are pairwise coprime; if x and y satisfy all three congruences, then 3, 5, 7 all divide $x - y$, hence $105 \mid x - y$.

Solution: A11

Polynomial division by the monic polynomial $x^2 + 1$ gives

$$f(x) = q(x)(x^2 + 1) + (a + bx)$$

with remainder degree < 2 . Thus each quotient class has a representative $a + bx$. In the quotient,

$$x^2 + 1 = 0,$$

so the class of x behaves as an element whose square is -1 . The map

$$\mathbb{R}[x]/(x^2 + 1) \rightarrow \mathbb{C}, \quad a + bx \mapsto a + bi$$

is in fact an isomorphism.

Solution: A12

If $J \triangleleft R$ contains I , then J/I is an ideal of R/I . Conversely, if $K \triangleleft R/I$, then $\pi^{-1}(K)$ is an ideal of R containing $\ker \pi = I$. These constructions are inverse: $\pi^{-1}(J/I) = J$, and $\pi(\pi^{-1}(K)) = K$ because π is surjective. Inclusion is preserved. This is the correspondence theorem for ideals.

A.4 Solutions to field and Galois exercises**Solution: G1**

$\sqrt{5}$ is a root of $x^2 - 5$, which is irreducible over $\mathbb{Q}$ because it has no rational root. Hence the minimal polynomial has degree 2 and

$$[\mathbb{Q}(\sqrt{5}) : \mathbb{Q}] = 2.$$

For the inverse, rationalize:

$$\frac{1}{3 + \sqrt{5}} = \frac{3 - \sqrt{5}}{(3 + \sqrt{5})(3 - \sqrt{5})} = \frac{3 - \sqrt{5}}{4}.$$

Thus the coordinates in the basis $\{1, \sqrt{5}\}$ are $3/4$ and $-1/4$.

Solution: G2

We have $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. We claim $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$. If $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, squaring gives

$$3 = a^2 + 2b^2 + 2ab\sqrt{2}.$$

Thus $ab = 0$. If $b = 0$, then $a^2 = 3$, impossible in $\mathbb{Q}$; if $a = 0$, then $2b^2 = 3$, likewise impossible. Hence the minimal polynomial of $\sqrt{3}$ over $\mathbb{Q}(\sqrt{2})$ still has degree 2. By the tower law,

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 2 \cdot 2 = 4.$$

Solution: G3

The calculation

$$\alpha^2 = 5 + 2\sqrt{6}, \quad (\alpha^2 - 5)^2 = 24$$

gives $\alpha^4 - 10\alpha^2 + 1 = 0$. To prove minimality, note

$$\alpha^{-1} = \sqrt{3} - \sqrt{2},$$

so

$$\sqrt{3} = \frac{\alpha + \alpha^{-1}}{2}, \quad \sqrt{2} = \frac{\alpha - \alpha^{-1}}{2}.$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which has degree 4 by G2. Therefore the minimal polynomial of α has degree 4, and the displayed monic degree-4 polynomial must be it.

Solution: G4

Define evaluation

$$\phi : \mathbb{Q}[x] \rightarrow \mathbb{Q}(\sqrt{2}), \quad f(x) \mapsto f(\sqrt{2}).$$

The image contains $\mathbb{Q}$ and $\sqrt{2}$, so it is all of $\mathbb{Q}(\sqrt{2})$. Since $x^2 - 2$ is the minimal polynomial of $\sqrt{2}$, the kernel is $(x^2 - 2)$. The first isomorphism theorem for rings gives

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}).$$

Concretely, every class has a representative $a + bx$, which maps to $a + b\sqrt{2}$.

Solution: G5

For $f = x^4 - 2$, $f' = 4x^3$. A common root would have to satisfy $x = 0$ and $x^4 = 2$, impossible, so $\gcd(f, f') = 1$ and f has no repeated roots.

In $\mathbb{F}_p(t)[x]$,

$$g'(x) = px^{p-1} = 0$$

because the characteristic is p . Thus the derivative test can fail dramatically: a nonconstant polynomial may have zero derivative. Indeed, in a suitable extension, $x^p - t = (x - \alpha)^p$ when $\alpha^p = t$, so the unique root occurs with multiplicity p .

Solution: G6

Let $\alpha = 2^{1/4}$. The roots of $x^4 - 2$ are

$$\alpha, -\alpha, i\alpha, -i\alpha.$$

Thus the splitting field is $L = \mathbb{Q}(\alpha, i)$. Eisenstein at 2 shows $x^4 - 2$ is irreducible, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$ and $i \notin \mathbb{R}$, $i \notin \mathbb{Q}(\alpha)$; adjoining i has degree 2. Hence in fact

$$[L : \mathbb{Q}] = 8,$$

which is stronger than the requested upper bound.

Solution: G7

Let $m(x) = \sum a_i x^i$ be the minimal polynomial of α over K . Since σ fixes K ,

$$m(\sigma(\alpha)) = \sum a_i \sigma(\alpha)^i = \sigma\left(\sum a_i \alpha^i\right) = \sigma(0) = 0.$$

Thus $\sigma(\alpha)$ is a root of m . Its own minimal polynomial divides m . Applying the same argument to σ^{-1} shows the minimal polynomial of α divides that of $\sigma(\alpha)$. Since both are monic, they are equal.

Solution: G8

In characteristic p ,

$$(a + b)^p = \sum_{k=0}^p \binom{p}{k} a^k b^{p-k} = a^p + b^p,$$

because $p \mid \binom{p}{k}$ for $0 < k < p$, so all middle coefficients vanish in the field. Also $(ab)^p = a^p b^p$ and $1^p = 1$. Thus Frobenius preserves addition, multiplication, and 1. The characteristic assumption is used exactly when the binomial coefficients divisible by p are set equal to zero.

Solution: G9

The multiplicative group $\mathbb{F}_{p^n}^\times$ has order $p^n - 1$. Hence for nonzero a ,

$$a^{p^n-1} = 1,$$

so $a^{p^n} = a$; the equation is also true for $a = 0$. Therefore every field element is a root of $x^{p^n} - x$. The polynomial has degree p^n and already has p^n roots, so these are all its roots. They are distinct because its derivative is

$$p^n x^{p^n-1} - 1 = -1,$$

which has no common root with the polynomial.

Solution: G10

Every independent sign choice

$$\sqrt{2} \mapsto \pm\sqrt{2}, \quad \sqrt{3} \mapsto \pm\sqrt{3}$$

preserves all rational algebraic relations, giving four automorphisms. Since the extension has degree 4, these are all automorphisms, and

$$\text{Gal}(L/\mathbb{Q}) \cong C_2 \times C_2.$$

Let σ negate $\sqrt{2}$ and fix $\sqrt{3}$. An element has the form

$$a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}.$$

Applying σ gives $a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$. Equality with the original forces $b = d = 0$, so the fixed field is $\mathbb{Q}(\sqrt{3})$.

Solution: G11

ζ_5 satisfies $x^5 - 1 = 0$. Any $\mathbb{Q}$ -automorphism preserves this polynomial, so it sends ζ_5 to a fifth root of unity. It also preserves multiplicative order: if $\sigma(\zeta_5)$ had order $d < 5$, then applying σ^{-1} would imply $\zeta_5^d = 1$. Hence the image is primitive. The primitive fifth roots are $\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4$, so

there are at most four choices for an automorphism, since an automorphism of $\mathbb{Q}(\zeta_5)$ is determined by the image of the generator ζ_5 .

Solution: G12

Eisenstein at 2 makes $x^3 - 2$ irreducible, so for $\alpha = \sqrt[3]{2}$,

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3.$$

The other roots are $\zeta\alpha$ and $\zeta^2\alpha$, where ζ is a primitive cube root of unity. The splitting field is $L = \mathbb{Q}(\alpha, \zeta)$. Since $\mathbb{Q}(\alpha)$ is real but ζ is not, $\zeta \notin \mathbb{Q}(\alpha)$, and its quadratic polynomial $x^2 + x + 1$ gives

$$[L : \mathbb{Q}] = 6.$$

The extension is Galois, so $|\text{Gal}(L/\mathbb{Q})| = 6$. Its action on the three roots gives an injective homomorphism into S_3 . Since S_3 also has order 6, the injection is an isomorphism.

A.5 Solutions to real-analysis exercises

Solution: R1

We have

$$\left| \frac{2n+1}{n+3} - 2 \right| = \left| \frac{-5}{n+3} \right| = \frac{5}{n+3}.$$

Given $\varepsilon > 0$, choose $N > 5/\varepsilon$. Then for $n \geq N$,

$$\frac{5}{n+3} \leq \frac{5}{n} \leq \frac{5}{N} < \varepsilon.$$

Hence $(2n+1)/(n+3) \rightarrow 2$.

Solution: R2

Suppose $a_n \rightarrow L$. Given $\varepsilon > 0$, choose N so that $|a_n - L| < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$|a_n - a_m| \leq |a_n - L| + |a_m - L| < \varepsilon.$$

Thus (a_n) is Cauchy.

Solution: R3

The alternating harmonic series is conditionally convergent: the terms $1/n$ decrease to 0, so the alternating-series test gives convergence, while $\sum 1/n$ diverges.

For $\sum 1/[n(\log n)^2]$, the integral test gives

$$\int_2^\infty \frac{dx}{x(\log x)^2} = \left[-\frac{1}{\log x} \right]_2^\infty = \frac{1}{\log 2} < \infty.$$

Thus it converges absolutely.

For $a_n = n!/n^n$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1} \right)^n \rightarrow e^{-1} < 1,$$

so the ratio test gives absolute convergence.

Solution: R4

For $r \neq 1$,

$$(1-r)(1+r+\dots+r^N) = 1-r^{N+1},$$

so

$$\sum_{n=0}^N r^n = \frac{1-r^{N+1}}{1-r}.$$

If $|r| < 1$, then $r^{N+1} \rightarrow 0$, giving $\sum_{n=0}^{\infty} r^n = 1/(1-r)$. The tail after N is

$$\sum_{n=N+1}^{\infty} r^n = \frac{r^{N+1}}{1-r},$$

so in absolute value it is at most $|r|^{N+1}/(1-|r|)$.

Solution: R5

On $[0, 1]$, the pointwise limit is 0 for $x < 1$ and 1 at $x = 1$, and convergence is not uniform because the sup error stays at least $1/2$ (choose $x_n = 2^{-1/n}$).

On $[0, a]$ with $a < 1$,

$$\sup_{x \in [0, a]} |x^n| = a^n \rightarrow 0.$$

Thus convergence to 0 is uniform. The difference is that the smaller interval stays a fixed positive distance away from the problematic boundary point 1.

Solution: R6

For $0 \leq x \leq 1$,

$$0 \leq \frac{x}{n+x} \leq \frac{x}{n} \leq \frac{1}{n}.$$

Hence

$$\|f_n\|_{\infty} \leq \frac{1}{n} \rightarrow 0,$$

so $f_n \rightarrow 0$ uniformly. In fact f_n is increasing in x , so the exact supremum is $1/(n+1)$.

Solution: R7

For every x ,

$$\left| \frac{\sin(nx)}{n^2} \right| \leq \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges, the Weierstrass M-test gives uniform and absolute convergence on all of $\mathbb{R}$.

Solution: R8

On $[0, 1]$,

$$\left| \frac{x^n}{n^2} \right| \leq \frac{1}{n^2},$$

and $\sum 1/n^2$ converges. Hence the function series converges uniformly by the M-test. Uniform convergence permits termwise integration:

$$\int_0^1 \sum_{n=1}^{\infty} \frac{x^n}{n^2} dx = \sum_{n=1}^{\infty} \frac{1}{n^2} \int_0^1 x^n dx = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}.$$

Solution: R9

Given $\varepsilon > 0$, choose N such that $|f_N(x) - f(x)| < \varepsilon/3$ for every x . Continuity of f_N at x_0 gives $\delta > 0$ such that $|x - x_0| < \delta$ implies $|f_N(x) - f_N(x_0)| < \varepsilon/3$. Then

$$|f(x) - f(x_0)| \leq |f - f_N|(x) + |f_N(x) - f_N(x_0)| + |f_N - f|(x_0) < \varepsilon.$$

Thus f is continuous.

Solution: R10

For all x ,

$$|f_n(x)| = \frac{|\sin(nx)|}{n} \leq \frac{1}{n},$$

so $f_n \rightarrow 0$ uniformly. But

$$f'_n(x) = \cos(nx), \quad f'_n(0) = 1$$

for every n , whereas the derivative of the uniform limit 0 is 0. Thus uniform convergence of functions alone does not justify passage of the limit through differentiation.

Solution: R11

For $\sum n^3 z^n$, $|a_n|^{1/n} = n^{3/n} \rightarrow 1$, so $R = 1$.

For $\sum z^n/n!$, the ratio test gives

$$\left| \frac{z^{n+1}/(n+1)!}{z^n/n!} \right| = \frac{|z|}{n+1} \rightarrow 0$$

for every z , so $R = \infty$.

For $\sum z^n/n$, $|a_n|^{1/n} = n^{-1/n} \rightarrow 1$, so $R = 1$. At $z = 1$ it is the harmonic series and diverges. At $z = -1$ it is the alternating harmonic series and converges conditionally.

Solution: R12

Fix $r < R$. Choose ρ with $r < \rho < R$. Since the power series converges absolutely at $z = \rho$, the numerical series

$$\sum |a_n| \rho^n$$

converges. For $|z| \leq r$,

$$|a_n z^n| \leq |a_n| r^n \leq |a_n| \rho^n.$$

The M-test therefore gives uniform convergence on $|z| \leq r$.

A.6 Solutions to complex-analysis exercises**Solution: C1**

Write

$$f(z) = z^2 \bar{z} = z|z|^2 = (x + iy)(x^2 + y^2),$$

so

$$u = x(x^2 + y^2) = x^3 + xy^2, \quad v = y(x^2 + y^2) = x^2y + y^3.$$

Then

$$u_x = 3x^2 + y^2, \quad u_y = 2xy, \quad v_x = 2xy, \quad v_y = x^2 + 3y^2.$$

Cauchy–Riemann requires $u_x = v_y$ and $u_y = -v_x$, giving

$$3x^2 + y^2 = x^2 + 3y^2 \Rightarrow x^2 = y^2,$$

and

$$2xy = -2xy \Rightarrow xy = 0.$$

Together these force $x = y = 0$. The partial derivatives are continuous, so the CR equations imply differentiability there. Thus f is complex differentiable only at 0 (but not holomorphic on any neighborhood of 0).

Solution: C2

Parametrize $z = Re^{it}$, $0 \leq t \leq 2\pi$. Then $dz = iRe^{it} dt$, so

$$\int_{|z|=R} \frac{dz}{z} = \int_0^{2\pi} \frac{iRe^{it}}{Re^{it}} dt = \int_0^{2\pi} i dt = 2\pi i.$$

Solution: C3

Cauchy's formula applies because e^z is entire and 1 lies inside the positively oriented circle $|z| = 2$. Therefore

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz = e^1 = e.$$

Solution: C4

The derivative form of Cauchy's formula says

$$\int_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(a).$$

Here $n = 2$, $f = e^z$, $a = 1$. Hence

$$\int_{|z|=2} \frac{e^z}{(z-1)^3} dz = \frac{2\pi i}{2!} e = \pi i e.$$

Solution: C5

Fix $z_0 \in \mathbb{C}$. On $|z - z_0| = R$,

$$|z| \leq |z_0| + R,$$

so $|f(z)| \leq A + B(|z_0| + R) =: M_R$. Cauchy's estimate gives for the second derivative

$$|f''(z_0)| \leq \frac{2M_R}{R^2} = \frac{2(A + B|z_0|)}{R^2} + \frac{2B}{R} \rightarrow 0.$$

Thus $f''(z_0) = 0$. Since z_0 was arbitrary, $f'' \equiv 0$, so $f(z) = az + b$.

Solution: C6

The exponential series gives

$$e^z - 1 = z + \frac{z^2}{2!} + \cdots,$$

so for $z \neq 0$,

$$\frac{e^z - 1}{z} = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \dots$$

Thus the limit as $z \rightarrow 0$ is 1. Defining $f(0) = 1$ removes the singularity and gives an entire power-series representation.

Solution: C7

For $1/[z(z-1)]$, at $z = 0$,

$$\text{Res} = -1,$$

and at $z = 1$,

$$\text{Res} = 1.$$

For $e^z/(z^2 + 1) = e^z/[(z-i)(z+i)]$, the residues are

$$\text{Res}(f; i) = \frac{e^i}{2i}, \quad \text{Res}(f; -i) = \frac{e^{-i}}{-2i}.$$

Equivalently use $g(a)/h'(a)$ with $h(z) = z^2 + 1$.

Solution: C8

The poles 0, 1, 2 are simple and lie inside. Their residues are

$$\frac{1}{2}, \quad -2, \quad \frac{5}{2},$$

which sum to 1. Hence the integral is $2\pi i$.

Solution: C9

For a pole of order 3,

$$\text{Res}\left(\frac{e^z}{(z-1)^3}; 1\right) = \frac{1}{2!} \frac{d^2}{dz^2} e^z \Big|_{z=1} = \frac{e}{2}.$$

Equivalently, expand $e^z = e e^{z-1} = e[1 + (z-1) + (z-1)^2/2 + \dots]$ and read off the coefficient of $(z-1)^{-1}$ after division by $(z-1)^3$.

Solution: C10

If $f(z) = (z-a)^m g(z)$ with $g(a) \neq 0$, then

$$\frac{f'}{f} = \frac{m(z-a)^{m-1}g + (z-a)^m g'}{(z-a)^m g} = \frac{m}{z-a} + \frac{g'}{g}.$$

The second term is holomorphic near a , so the residue is m . If f has a pole of order m , write $f = (z-a)^{-m}g$; the same calculation gives residue $-m$.

Solution: C11

Let $h = f - g$. Then h is holomorphic and has zeros at the distinct points $z_n \rightarrow z_0 \in U$. If h were not identically zero, its zeros would be isolated, contradicting the accumulation of zeros at the interior point z_0 . Hence $h \equiv 0$, so $f \equiv g$.

Solution: C12

On $|z| \leq R$,

$$\left| \frac{z^n}{n!} \right| \leq \frac{R^n}{n!}.$$

The numerical series $\sum R^n/n!$ converges by the ratio test. Hence the M-test gives uniform convergence on every closed disk. Each partial sum is a polynomial and therefore holomorphic. Local uniform convergence of holomorphic functions implies the limit is holomorphic on all of $\mathbb{C}$; thus it is entire.

A.7 Solutions to synthesis exercises**Solution: S1**

The set μ_n is closed under multiplication and inverses because if $z^n = w^n = 1$, then $(zw)^n = 1$ and $(z^{-1})^n = 1$. It contains 1, so it is a group. The roots of $z^n = 1$ are exactly

$$\zeta_n^k = e^{2\pi i k/n}, \quad 0 \leq k < n,$$

so ζ_n generates all n elements. The order of ζ_n^a is the least positive m such that $n \mid am$, namely

$$\frac{n}{\gcd(a, n)}.$$

Thus it has order n exactly when $\gcd(a, n) = 1$.

Solution: S2

The polynomial $x^3 + x + 1$ has no root in $\mathbb{F}_2$: at 0 it is 1, and at 1 it is $1 + 1 + 1 = 1$. A cubic over a field is reducible iff it has a root, so it is irreducible. If α is a root, then the Frobenius conjugates are

$$\alpha, \quad \alpha^2, \quad \alpha^4.$$

The next iterate is α^8 . Every element of $\mathbb{F}_8$ satisfies $x^8 = x$, so $\alpha^8 = \alpha$. Since the minimal polynomial has degree 3, the orbit cannot return earlier; otherwise α would lie in a proper smaller finite subfield and have smaller degree over $\mathbb{F}_2$.

Solution: S3

Using geometric series,

$$(1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1} = \left(\sum_{a \geq 0} 2^{-as} \right) \left(\sum_{b \geq 0} 3^{-bs} \right) = \sum_{a, b \geq 0} (2^a 3^b)^{-s}.$$

Exactly the positive integers whose prime factors all lie in $\{2, 3\}$ occur, each exactly once by unique factorization.

Solution: S4

Let P_y be the finite set of primes $p \leq y$. Then

$$\prod_{p \leq y} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ p|n \Rightarrow p \leq y}} n^{-s}$$

by expanding finite products of convergent geometric series and using unique factorization. As $y \rightarrow \infty$, the right-hand side increases monotonically to $\sum_{n \geq 1} n^{-s}$ because every positive integer has finitely many prime factors. Since $s > 1$, this series converges. Therefore

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \lim_{y \rightarrow \infty} \prod_{p \leq y} (1 - p^{-s})^{-1},$$

which is the Euler product. Positivity makes the limiting argument especially transparent; in the complex half-plane $\Re(s) > 1$, absolute convergence plays the corresponding role.

Solution: S5

If α is a root of $f \in K[x]$ and $\sigma \in \text{Gal}(L/K)$, then

$$f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0,$$

so the roots are stable under the group action. If f is irreducible and splits in L , the Galois group acts transitively on its roots: for any two roots α, β , the K -isomorphism $K(\alpha) \rightarrow K(\beta)$ sending α to β extends to an automorphism of a normal splitting field. Thus the roots form one orbit.

Solution: S6

If $y = g \cdot x$, then by assumption

$$\sigma_y = g \sigma_x g^{-1}.$$

Thus σ_y is conjugate to σ_x . Any two choices in the same orbit differ by some group element g , so they produce elements in the same conjugacy class. Therefore the class $[\sigma_x]$ depends only on the orbit, not on the chosen representative.

Solution: S7

Take logarithms of the finite product:

$$\log F_P(s) = - \sum_{p \in P} \log(1 - p^{-s}).$$

Differentiate:

$$\frac{F'_P}{F_P} = - \sum_{p \in P} \frac{(\log p) p^{-s}}{1 - p^{-s}}.$$

Therefore

$$-\frac{F'_P}{F_P} = \sum_{p \in P} (\log p) \frac{p^{-s}}{1 - p^{-s}} = \sum_{p \in P} \sum_{k \geq 1} (\log p) p^{-ks}.$$

The logarithmic derivative has converted a product indexed by primes into an additive sum indexed by prime powers.

Solution: S8

Many correct graphs are possible. One representative chain is:

ideal $\rightarrow$ quotient ring $\rightarrow$ field extension $\rightarrow$ minimal polynomial $\rightarrow$ Galois automorphism $\rightarrow$ Frobenius,

where ideals make ring quotients well-defined; quotienting $K[x]$ by an irreducible minimal polynomial constructs a field extension; minimal polynomials determine conjugate roots; Galois automorphisms permute those roots; and over finite fields Frobenius is a canonical automorphism.

On the analytic side,

uniform convergence $\rightarrow$ holomorphic limits $\rightarrow$ residues,

while

unique factorization $\rightarrow$ Euler product $\rightarrow$ logarithmic derivative $\rightarrow$ prime-power sums.

The important feature is not the exact drawing but that every arrow is supported by a mathematical sentence rather than visual association alone.

A.8 Solutions and marking guide for the final readiness test

Solution: F1

Normal subgroup: $N \leq G$ with $gNg^{-1} = N$ for every $g \in G$. Example: $A_3 \triangleleft S_3$.

Prime ideal: a proper ideal $\mathfrak{p}$ such that $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Example: $(5) \subset \mathbb{Z}$.

Minimal polynomial: for algebraic α over K , the unique monic polynomial of least positive degree in $K[x]$ vanishing at α . Example: $x^2 - 2$ for $\sqrt{2}$ over $\mathbb{Q}$.

Galois extension: a finite extension that is normal and separable. Example: $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$.

Uniform convergence: $f_n \rightarrow f$ uniformly on X if for every $\varepsilon > 0$ there is N such that for all $n \geq N$ and all $x \in X$, $|f_n(x) - f(x)| < \varepsilon$. Example: $x^n \rightarrow 0$ uniformly on $[0, 1/2]$.

Residue: the coefficient of $(z - a)^{-1}$ in the Laurent expansion of a meromorphic function about an isolated singularity a . Example: $\text{Res}(1/z; 0) = 1$.

For full marks, the formal statement and example both need to be correct; intuition is a bonus sign of robust recall.

Solution: F2

If $\mathfrak{m}$ is maximal and $a \notin \mathfrak{m}$, then $\mathfrak{m} + (a) = R$, so $1 = m + ra$. Modulo $\mathfrak{m}$, the class of r inverts the nonzero class of a . Thus $R/\mathfrak{m}$ is a field. Conversely, if the quotient is a field, an ideal strictly between $\mathfrak{m}$ and R would descend to a nonzero proper ideal of the field, impossible.

The prime criterion uses “integral domain” because primality prohibits zero divisors in the quotient, whereas maximality forces every nonzero quotient class to be invertible, the stronger field property.

Solution: F3

The conjugacy classes are $\{e\}$, the three transpositions, and the two 3-cycles. Under conjugation, orbit size equals class size. Stabilizers are centralizers.

For e , the stabilizer is all of S_3 , size 6, so $1 \cdot 6 = 6$.

For a transposition such as (12) , the centralizer is $\{e, (12)\}$, size 2, so orbit size 3 and $3 \cdot 2 = 6$.

For a 3-cycle such as (123) , the centralizer is its cyclic subgroup $\{e, (123), (132)\}$, size 3, so orbit size 2 and $2 \cdot 3 = 6$.

Solution: F4

As before, $\alpha = \sqrt{2} + \sqrt{3}$ satisfies

$$\alpha^4 - 10\alpha^2 + 1 = 0.$$

Since $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, both radicals lie in $\mathbb{Q}(\alpha)$, so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. This field has degree 4 by the tower law and the fact $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$. Hence the displayed degree-4 polynomial is minimal and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$.

Solution: F5

In characteristic p , Frobenius $x \mapsto x^p$ preserves sums and products because the intermediate binomial coefficients vanish modulo p . Iterating r times gives $x \mapsto x^q$ for $q = p^r$. On the finite field $\mathbb{F}_{q^n}$ this is an automorphism. Every $a \in \mathbb{F}_q$ satisfies $a^q = a$, so it fixes $\mathbb{F}_q$; every $x \in \mathbb{F}_{q^n}$ satisfies $x^{q^n} = x$, so the n th iterate is the identity.

This is a precursor to global Frobenius because later an unramified prime determines an automorphism through its induced action $x \mapsto x^{N\mathfrak{p}}$ on a finite residue field.

Solution: F6

The four independent sign changes of $\sqrt{2}$ and $\sqrt{3}$ give

$$G \cong C_2 \times C_2.$$

Its three subgroups of order 2 are generated respectively by: negating $\sqrt{2}$ only, negating $\sqrt{3}$ only, and negating both. Their fixed fields are respectively

$$\mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{6}).$$

These are exactly the three quadratic intermediate fields. The full group fixes $\mathbb{Q}$, and the trivial subgroup fixes L .

Solution: F7

The pointwise limit of x^n on $[0, 1]$ is 0 on $[0, 1)$ and 1 at 1. It is not uniform because for every n there are points $x < 1$ arbitrarily close to 1 with x^n bounded away from 0; for instance $x_n = 2^{-1/n}$ gives error $1/2$.

The quantifier issue is that pointwise convergence allows N to depend on x , but uniform convergence demands one N valid for all x . On $[0, a]$ with $a < 1$,

$$\sup_{0 \leq x \leq a} x^n = a^n \rightarrow 0,$$

so the convergence is uniform.

Solution: F8

For $a_n = n^2/3^n$,

$$\limsup |a_n|^{1/n} = \frac{1}{3},$$

so $R = 3$. On $|z| = 3$, the n th term has magnitude n^2 and therefore does not tend to zero, so the series diverges everywhere on the boundary.

On any smaller disk $|z| \leq r < 3$, choose ρ with $r < \rho < 3$. Absolute convergence at ρ provides a summable majorant $|a_n|\rho^n$, and the M-test gives uniform convergence on the r -disk.

Solution: F9

For the first integral, Cauchy's derivative formula with $f = e^z$ and $a = 1$ gives

$$\int_{|z|=2} \frac{e^z}{(z-1)^2} dz = 2\pi i f'(1) = 2\pi i e.$$

For the second, the residues at 0, 1, 2 sum to 1, so the residue theorem gives $2\pi i$.

Solution: F10

Unique factorization gives, for $\Re(s) > 1$,

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_p (1 - p^{-s})^{-1}.$$

The product is possible because each integer corresponds uniquely to a choice of prime exponents. Taking logarithmic derivatives converts products into sums:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{(\log p)p^{-s}}{1 - p^{-s}} = \sum_p \sum_{k \geq 1} (\log p)p^{-ks}.$$

Equivalently, using the von Mangoldt function $\Lambda(n)$, defined by $\Lambda(p^k) = \log p$ and 0 otherwise,

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}.$$

Thus multiplicative information about all integers is packaged in an Euler product, and logarithmic differentiation extracts an additive series concentrated exactly on prime powers. Later analytic information about the left side is converted back into distributional information about primes.

Appendix B

Compact recall deck

This appendix is not intended for first learning. Use it for spaced recall after you have worked through the chapters. Cover the right-hand column and reconstruct it from the term on the left; then reverse the exercise and try to name the term from the description.

Term	Definition / memory cue
Group	A set with associative composition, identity, and inverses. Mental model: reversible operations or symmetries.
Subgroup	A subset that is itself a group. Efficient test: nonempty and $ab^{-1} \in H$ whenever $a, b \in H$.
Normal subgroup	$N \leq G$ with $gNg^{-1} = N$ for all g . Exactly the condition making multiplication of cosets well-defined.
Homomorphism	A map preserving the operation. Kernel = information made invisible; image = structure that survives.
Group action	A homomorphism $G \rightarrow \text{Sym}(X)$, written $g \cdot x$. Turns abstract group elements into actual symmetries of objects.
Orbit	$\text{Orb}(x) = \{g \cdot x : g \in G\}$. Where x can move.
Stabilizer	$\text{Stab}(x) = \{g : g \cdot x = x\}$. Symmetries that leave x fixed.
Ideal	Additive subgroup I of a ring with $rI \subseteq I$ for every $r \in R$. Exactly what is needed for ring quotients.
Prime ideal	Proper $\mathfrak{p}$ with $ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Equivalent to $R/\mathfrak{p}$ being a domain.
Maximal ideal	Proper $\mathfrak{m}$ with no proper ideal strictly between $\mathfrak{m}$ and R . Equivalent to $R/\mathfrak{m}$ being a field.
Comaximal ideals	$I + J = R$, equivalently $1 = a + b$ with $a \in I, b \in J$. Means congruence conditions are independent enough for CRT.
Field extension	Inclusion of fields $K \subseteq L$; degree $[L : K] = \dim_K L$.
Algebraic element	α satisfies a nonzero polynomial over the base field.
Minimal polynomial	Unique monic polynomial of least degree over K vanishing at α . Irreducible, and its degree equals $[K(\alpha) : K]$.
Splitting field	Smallest extension containing all roots of a polynomial.
Separable	Roots of the relevant irreducible polynomials are distinct.
Normal	If an irreducible polynomial over the base has one root in the extension, it has all of them there.
Galois extension	Finite, normal, and separable. Symmetries have full access to the complete set of conjugate roots.
Galois group	Base-field-fixing automorphisms $\text{Gal}(L/K)$. They permute roots while preserving algebraic relations.
Frobenius	In characteristic p , $x \mapsto x^p$; over $\mathbb{F}_{q^n}/\mathbb{F}_q$, $x \mapsto x^q$ generates the canonical cyclic Galois behavior.
Sequence convergence	$\forall \varepsilon > 0 \exists N \forall n \geq N : a_n - L < \varepsilon$. Every demanded accuracy is eventually met permanently.
Cauchy sequence	Late terms become arbitrarily close to one another. Detects convergence without naming a limit.

Term	Definition / memory cue
Absolute convergence	$\sum a_n < \infty$. Strong form of series convergence robust under rearrangement/-manipulation.
Pointwise convergence	For each x , one may choose a separate $N(x, \varepsilon)$.
Uniform convergence	One $N(\varepsilon)$ works simultaneously for every x . Equivalent to sup-norm error tending to zero.
M-test	$ f_n(x) \leq M_n$ and $\sum M_n < \infty$ imply uniform absolute convergence of $\sum f_n$.
Holomorphic	Complex differentiable at every point of an open set. Strong because the derivative limit must agree from every complex direction.
Cauchy formula	Interior values and derivatives of holomorphic functions are determined by contour integrals around them.
Zero of order m	$f(z) = (z - a)^m g(z)$ with $g(a) \neq 0$.
Pole of order m	$f(z) = (z - a)^{-m} g(z)$ with $g(a) \neq 0$.
Residue	Coefficient of $(z - a)^{-1}$ in the Laurent expansion. The only Laurent mode surviving one contour integration.
Euler product	Product over primes encoding unique factorization, e.g. $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ for $\Re(s) > 1$.
Logarithmic derivative	f'/f turns products into sums and zeros/poles into simple poles carrying multiplicities.

Appendix C

What comes next

Phase 1 begins systematic commutative algebra: Noetherian rings, localization, integral dependence, discrete valuation rings, and Dedekind domains. The concepts refreshed here should now be functional enough that those new structures can be learned for their own sake.

Do not require perfect recall before moving on. A good Phase 0 outcome is that a forgotten definition now feels *reconstructible*: you remember what structural problem it solves, so the formal clauses can be rebuilt rather than guessed.

Keep the following chain visible during the next phase:

quotients $\rightarrow$ prime ideals $\rightarrow$ localization $\rightarrow$ Dedekind domains $\rightarrow$ prime ideal factorization
--

That is the algebraic road on which the eventual Frobenius elements will travel.